Building Surveying Guide

PKNail Pro for Revit V1.3

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Introduction

The purpose of this manual is to introduce an individual to the concepts and practices of building surveying and the creation of BIM/Revit model utilizing building surveying techniques and PKNail Pro software from PointKnown.

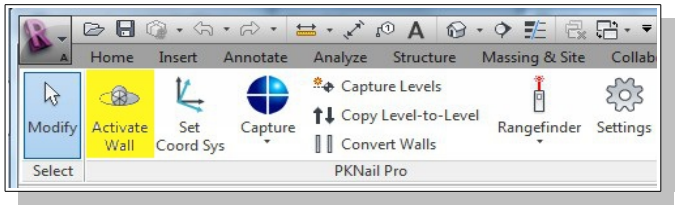
Whether it is a Tablet PC, a Notebook or Surface Tablet this guide assumes you will be running Autodesk Revit Architecture on a field workstation. Utilizing PKNail Pro will save time, increase accuracy and give you a product at the end of the day that can be utilized by anyone.

Techniques described in the manual range from generic surveying techniques, to using Revit, to using PKNail Pro with Revit in order to get the most out your day.

Project Scope

Crucial to the success of any project is creating the scope of the project with your client. What is the purpose of the model? What is the level of detail needed to get to this goal? Defining it can help you choose the right technology.

Activate Wall



PKNail Pro was developed by field professionals for field professionals and is intended to work as expected. So why mention this under 'activate wall', and why put it in the very front of the manual? Well this command will put our 'surveyor' on any wall, one created by PKNail Pro Software, a wall dragged by a user natively in Revit or one created in the office through a different procedure and in most cases will start or help you continue the process. So if you need to survey something, or switch ends of a surveyed wall, get to the other side of an end wall, simply click 'Activate Wall', select the location you want to be and get ready to go. Activate Wall will be discussed in more detail in the following sections, but it was important to mention right up front.

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Tools

You will be creating Revit models while you measure so in order to optimize that experience we have included needed and helpful tools.

- **Field Workstation**
  Must either have an internal Bluetooth radio or external USB Bluetooth Dongle for communication. As for horsepower and RAM, we have had plenty of success using sub $1000 systems. Remember this is a field/mobile workstation; detail work we usually leave for the office. Much of what we model in the field uses generic components which allows for lighter weight files

- **External Battery Pack / Adapters**
  We prefer an external battery pack because it can be used universally across laptops, phones, etc. but it requires a place where it can be 'stored' while connected to the laptop. Additionally, they can be plugged directly into the wall rather than needing a laptop to charge. Or use your brands specific laptop battery.

- **Disto Laser** (330i or Disto D8): See Exhibit A for Disto D8 Layout and Features
  Currently PKNail Pro uses the Disto Line or Laser Range Finders from Leica Geosystems. And these two models are Bluetooth enabled which is a necessity. The D8 has more on board functionality and is a muse for working outdoors, especially for it's 4X Zoom and Color screen for targeting.

- **Laptop Tripod or Mobile Cart**
  While carrying a Windows Tablet for short term use is okay we highly recommend the purchase of laptop tripod or laptop cart for use in the field. It provides a stable work surface, is mobile and saves you from needing Popeye like strength in carrying a tablet or laptop through out the day.

- **Reflective Target**
  At distance and working alone a reflective target either available through Leica or other suppliers allows you a larger target that can be seen from distance, allows for longer measurements, and is highly visible when the laser hits it. We typically affix these with Painter's tape to the side of wall since Painter's tape will quick release without marring a surface.

- **Sticky Notes**
  These provide a quick target, especially for interior walls and on outside corners that will allow you obtain an accurate measurement.

- **Painters Tape**

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Used for affixing targets. Quick release.

- **Measuring Tape**
  Cloth and Spring Steel
  Why both? Cloth allows you to wrap a column quickly for an accurate measurement and determine it diameters from its circumference.
  Calculation: Diameter = Circumference / ∏ or D = C / 3.1416

  See Exhibit B for Conversion Chart

  Additionally, you can find 'Diameter' Tape and/or Calipers that can quickly calculate a diameter of a column or, in the case of the calipers, determine a wall thickness.

- **Building Chalk / Marking Crayon**
  On occasion you will need to mark a wall with a target, more often than not when calculating an angle or arc. This can be done sometimes with a sticky note and an 'x' or simply using chalk or building crayon to mark an 'x'. However, not always possible to use something that can be easily erased or removed.

- **LED Flashlight / Wearable**
  Hands free option is a wearable LED head lamp or simply keep a flashlight in your bag

- **Extra Batteries**
  AA for the Disto, rechargeable with recharger best choice.

- **Snacks**
  Being well fed or at least not hungry keeps you sharp. Stay sharp. Bars and Trail Mix always a good option and last a while if not eaten.



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Disto Keypad and Defined Keys for PKNail Pro

The keypad of the Disto allows you to interact with PKNail Pro and Revit wirelessly. In order to perform its many functions the keys take on roles you should familiarize yourself with.



Commonly Used Disto Functions

The Disto has two modes one is its standard mode, the second is enabled by activating the 2nd Key. In standard mode you have access to a wide array of on board Disto functions, when the 2nd Key is pressed, when you have PKNail Pro activated, you will be accessing PKNail Pro commands.

To get the most out of both products, you can access the Leica D8 Manual, as well as view some of the most commonly used standard modes outlined below.

Add and Subtract



To compile measurements using the '+' key or subtracting volumes using the ' – ' key, it works exactly like a calculator. Take a measurement by placing the range finder and pushing the red button, and then use the proper keys to add, subtract or get your final sum with the " = " button.

Digital Pointfinder



The Digital Pointfinder allows you to activate on board viewfinder on the color screen of the Disto. It has a 4x Digital Zoom so allows you to use the viewfinder to locate your target rather than tracing a red dot with the naked eye. This is very useful for outdoor work and when working over distances.



Selecting Tripod Mode

Minimum/Maximum Measurement

The Disto allows you to take minimum and maximum measurements of any given are by sweeping it across the area you want to measure. This comes in very used full when triangulating spaces, shooting inside corners, and taking diagonal measurements. To activate press and hold the red 'on' button until beeps. Sweep the red dot or laser across the surface or area you want to measure. The Disto will keep taking measurements until you press the 'on' button again to stop the process. Both maximum and minimum measurements will be displayed.

Tripod Mode

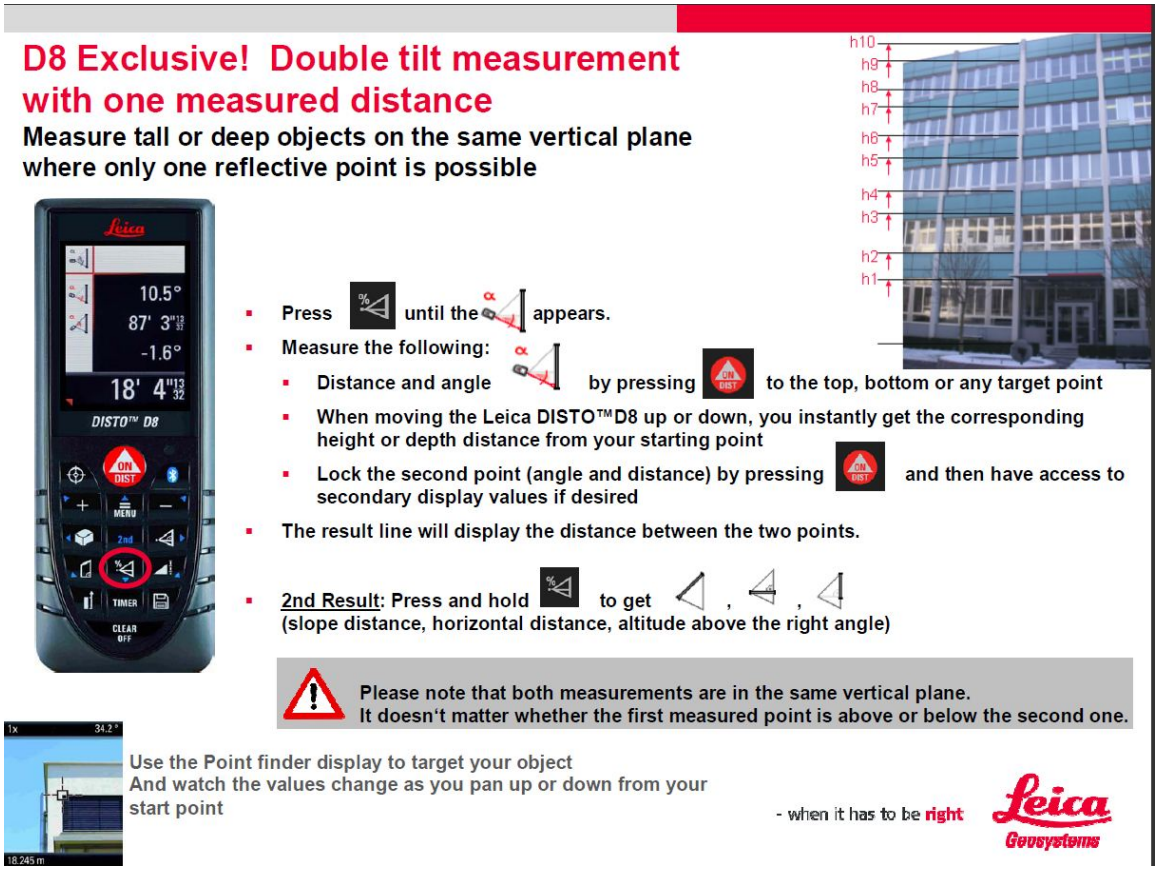


When mounting the Disto on a Tripod for indirect measurements you should put the Disto in Tripod mode for the most accurate measurements. By using Tripod Mode the Disto will calculate measurements based on the correct axis of the rangefinder. It is recommended to use the Leica Tripod Adapter for best results.

To enable Tripod Mode, hold down the Menu Button. Scroll and select the icon for the Tripod.
Do not forget to take the Disto out of Tripod mode when you are finished.

Finding Vertical Distances /Pythagorean Calculation 2

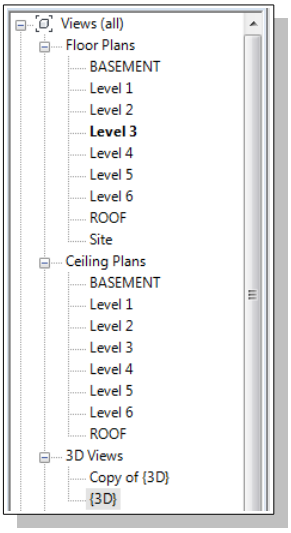
One of the most used functions when surveying the exterior of a building is finding vertical distances between objects.



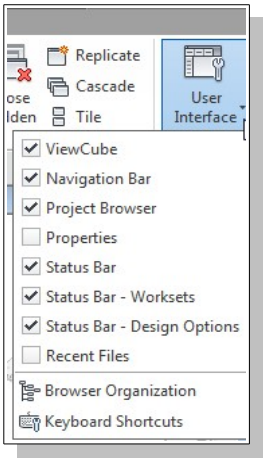
Common Revit Functions

While PKNail Pro enables you to create Revit objects it is helpful to understand Revit and have some commands at your disposal to assist in documenting an existing building. There is much more extensive help within Revit and its User Manual and this section intended only as a brief primer.

Switching Views

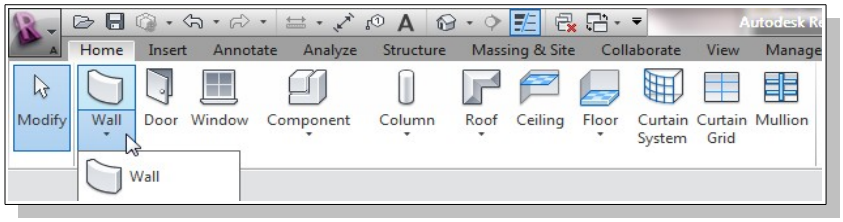


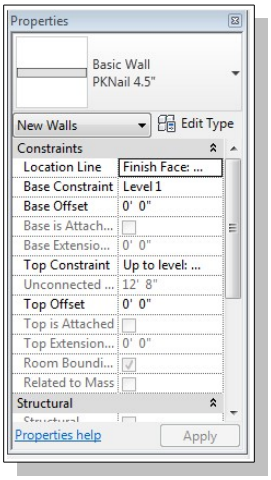
To switch views in Revit or open another view. Simply click on the Floor you want to view or work in inside the Project Browser. This is typically on the justified to the left side of the Revit Screen. If it is not open you can go to the 'View' Tab in the Revit Menu and to the far left is a selection called 'User Interface' and make sure the 'Project Browser' Item is selected.



Create a Wall

To create your own wall in Revit simply go to the 'Home' Tab in Revit and select 'Wall'

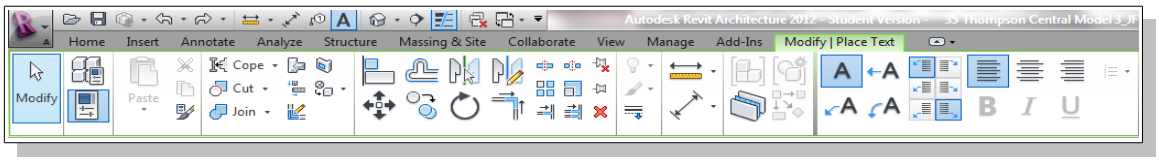




You are ready to place wall in the model. You can select what wall you want within the Properties Box, and select its location line. Basically, when documenting an existing building you want to choose either 'Finished Face Exterior' or 'Finished Face Interior'. Then click in the model where you want to place the wall and either enter the length you want it to be or drag it to the correct length. If the wall is justified incorrectly simply tap the space bar while you are dragging it to toggle between 'Location Lines'.

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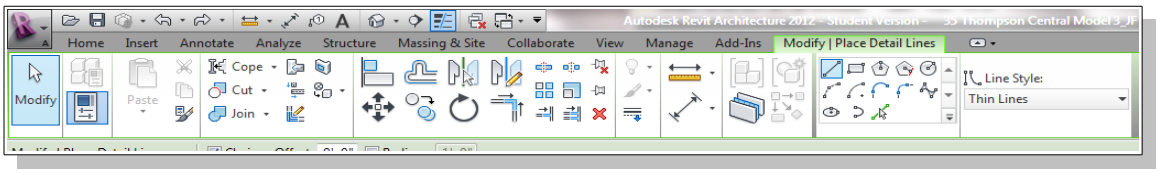
Creating Text or Annotations (tx)



To create an Annotation in the Model simply type 'tx' and a text cursor will appear on the screen, click on the model where you want to add the text and simply type the annotation. You can change the text size in the properties box.



Creating a Detail Line (dl)



To create a detail line, type 'dl' and then click where you want to start a detail line and then drag it in the correct direction and its correct length or enter the correct length. This is helpful if you want to mark something within the model.

You will also notice within the Detail Line Menu (which appears after you type 'dl' there are options to create circles and arcs.

Changing Wall Widths

To change a wall width, simply select the wall, and then select the correct wall from the 'Properties'. The wall will expand/contract from the initial surveyed side of the wall. So if you were to change a 4” wall to a 6” wall, it will expand away from the surveyed side. Consequently, the new wall width will have encroached into a surveyed space. If this was not intended you will need to move walls to account for this new wall thickness.

Move Command (mv)

Type 'mv' in order to move objects. You can click and drag the cursor over items or hold down the 'ctrl' key to click and add (+) items or the 'shift' key to subtract (-) objects. Note: Click and drag left selects all items within the pick box, click and drag right only selects objects who are entirely within the pick box.

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Building Types

To better understand surveying techniques it's best to understand the types you might encounter in the field. These types are described in more detail in another section. Important to understand is what buildings have in common and how you can utilize common features during surveying and data capture.

Common Building Features

Window Alignment

Most buildings were designed and created to reuse similar features and create symmetry within the design. For example, windows will typically 'stack' or align themselves vertically up the face of a building.

Vertical Transport / Cores

In bigger building the vertical transportation or cores will typically align and is the same throughout the building. For example, enclosed fire stairs and elevator shafts will align vertically through the building with the same construction methods. For example, the entire fire stair core will be built from CMU block.

PKNail Pro takes into account this design features by allowing you to stack windows, or 'copy' them from level to level as well as copy walls from level to level based on the information you have recorded with the software (See "Copy Function" for full documentation of this feature).

Columns and Column Grids

Columns whether in a timber framed or steel framed structure will typically align vertically through a building. This was done to transmit the vertical load of the floor to the structure below, this load would be transmitted through columns aligned vertically through the building to the footings. The grid once surveyed can be placed through out the building. A surveyor is initially limited to measuring to the face of a column (round or square) from surrounding structure. However, once a variety of columns are placed by this method, and measurements are taken to their center point one can start to deduce what the 'grid' is in the building.

Samples should be taken on every floor and assumptions should be verified.

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Survey Techniques

The start of any good surveying project is setting it up correctly and gathering critical information about a building. Typically that starts with structural information about the buildings, such as wall widths and building heights. When starting with PKNail Pro, you typically follow a Z axis workflow before placing your first wall. That is you document the number of Levels you will be surveying, plus Floor to Floor and Floor to Ceiling Heights, as well as placing the Typical Window Heights and Sill Heights by Level. While we will discuss that in depth later in the guide, for now, we will be discussing, generically, how to capture some basic information about a building.

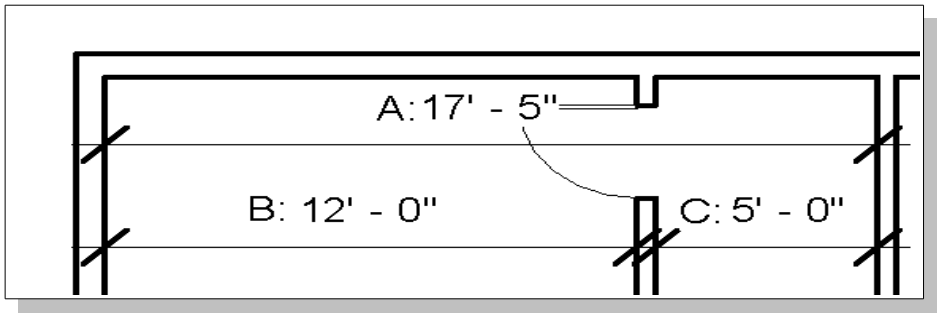
Subtracting Volumes

This is generally thought of taking an over all dimension and reducing it by defined measurements to find the measurement or dimension of an object or level.

Finding Wall Widths

By Subtracting Volumes

Wall widths are sometime easiest to ascertain by measuring the over all length of 2 rooms with an interior partition wall in the middle, and then reducing the overall length by the width in one and then reducing it again by the width of the other room leaving only the width of the interior partition wall. This is best implemented using a laser range finder and the 'subtract' function. A – B – C = Wall Width. While mathematically the equation could be written A – (B + C) we are working with Disto logic so take the over all between two rooms and then subtracting the two room volumes will leave you with the wall thickness.



In the above case, take an over all measurement with the Disto, 17'-5" (A) and then subtract 5'-0" (B) and 12'-0" which equals 5". Or 17'-5" - (12' + 5') = 5"

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Measuring

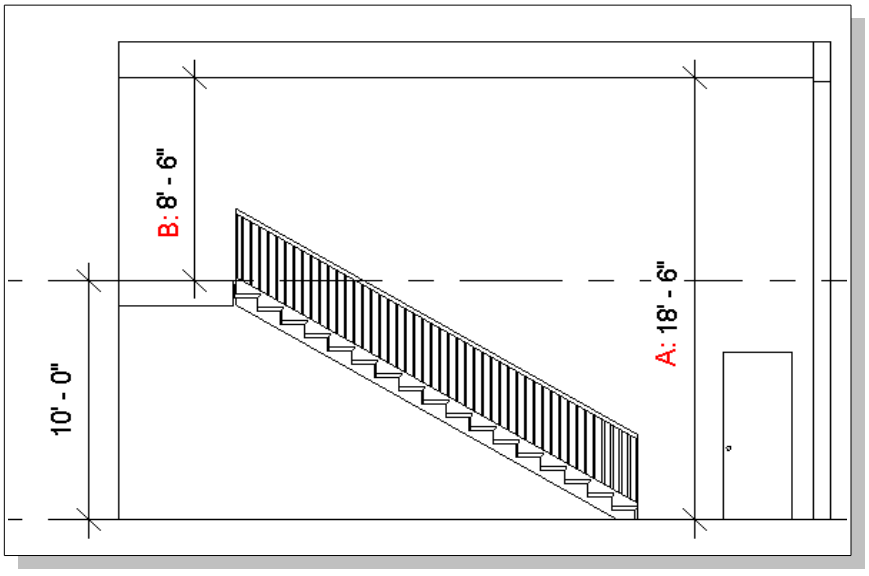
You can use a measuring tape to measure each component and/or the entire wall width. With farmed doorways do not forget to subtract the trim detail, either metal or wood from the over all width of the door jamb. There are also calipers available or that can be modified to measure wall widths easily. More often than not these are available at timber/tree/forestry product specialty shops.

Finding Level to Level Heights

Perhaps one of the most important parts of modeling and data collection on existing buildings is creating the correct levels for the model. The best way if available is to take a measurement directly from the finished floor of one level to the next. This is sometimes possible in stair cores with multiple levels. Secondly, like walls, is the ability to subtract or add volumes. For example, finding a common ceiling in a stairwell or like in the example below.

By Subtracting Volumes

This same principle can often be applied to finding different levels. For example, in the following illustration you can see how using common features, in this case, measuring to and utilizing a common ceiling that can allow you to determine the distance between levels. Where a surveyor can measure distance A, subtract distance B and determine the correct level to level height.



In this example the surveyor could measure to the common ceiling from Level 1 (18'-6”) and subtract the distance of the same ceiling to the next level (8'-6”) leaving the correct Level to Level Measurement of 10'-0”: 18'-6” - 8'-6” = 10'-0”

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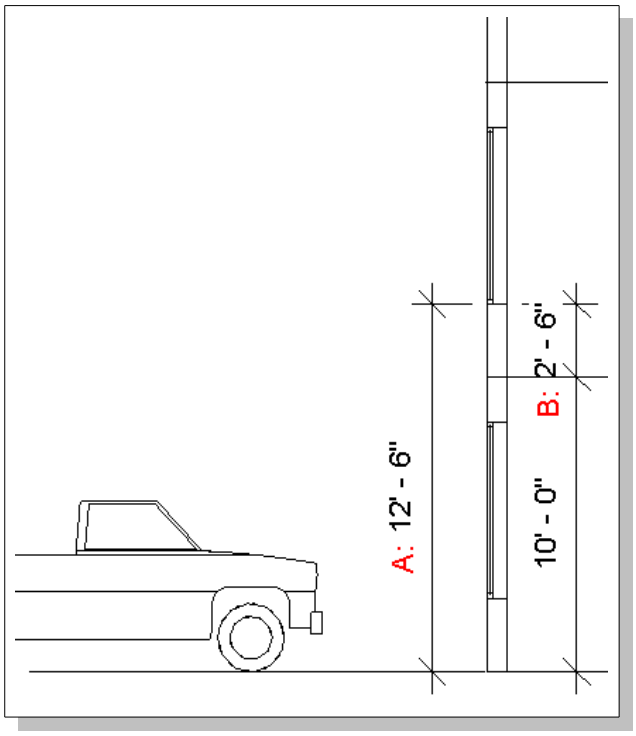
Measuring (Interior)

Many times in a building it is possible to get a clean laser shot from one level to another withing the stair core, or stair well. This is preferred when available.

Otherwise, you can measure risers on the stairs if needed. Typically stringers were cut or manufactured using exact dimensions, for example a 7” Rise. So counting risers and multiplying by a sampled riser measurement is one method, or measuring every riser with the Disto or measuring tape and adding them together.

Measuring (Exterior)

If you captured the sill heights correctly from the exterior you can then translate those measurements to the interior. This requires interfacing directly in Revit and drafting or adding Level Lines in an Elevation View. Initially, if you placed Windows from the Exterior Only you will have 2 Levels, the Base Level and the Roof/Top Level. To add additional levels you can go to an elevation view, take a measurement from the Window Sill down the Level you are on. For example, 2'-6”. A new level is 2'-6” below the Window Sill Height. Copy a Level to that position, rename it to the correct Level on move on to the other Levels using this method or one of the others already described.



In this example, the surveyor has found a sill height from the Exterior / Base Level: Level 1 to the Sill Height of the Window on the 2nd Level of 12'-6”.

If needed the surveyor can go inside and them measure down from the sill to Level 2 to determine the proper Level to Level Height.
A – B = C
12'-6” - 2'-6” = 10'- 0”

Level 2 Sill Height Exterior - Level 2 Sill Height Interior = Level 1 to Level 2 Distance.

*Note a Direct Measurement from Floor to Floor is preferred, and then typically using one of the interior methods. The elevation of the exterior sill and interior sill are most often different and this has to be accounted for if and when you are setting level heights with the exterior method.

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Finding Clean Lines / Overall Dimensions

It is important to understand the extent of the building you are surveying. Most important finding the overall exterior dimension and the over interior dimension provides you with the shell of the building and the typical exterior wall widths.

Almost all survey tools, no matter what technology, require finding clean lines of sight. When assessing a building or starting the survey work, walking the building first will offer insights and will present the surveyor the best options for starting the survey.

- Is the exterior free and clear of obstructions?
- Is a floor or free of interior wall partitions? Unoccupied?
- Can you use the roof to determine the outline of the building?
- Can you find an overall interior shot (interior face of an exterior wall to the opposite interior face of an exterior wall)?

Note: You can use the add/subtract functions of the laser meter to create an overall dimension if needed.

Using a Roof Line / Parapet

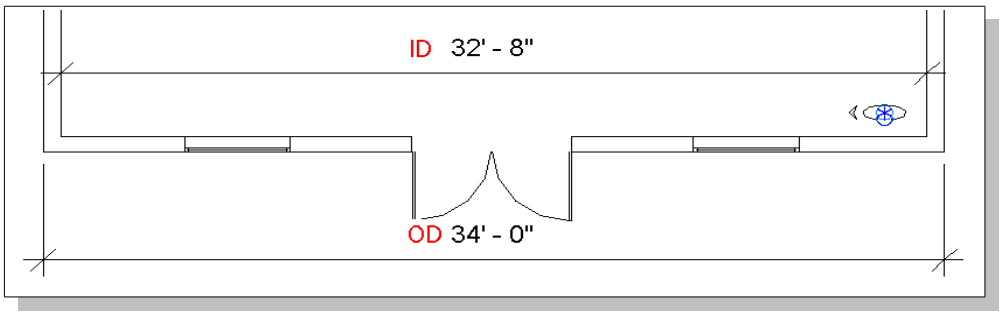
Depending on the site and type of building you may be able to get the dimensions and overall profile of a building from the roof.

Note: Use proper safety equipment and techniques.

Setting Extents

It is important to understand the extent of the building you are surveying and different surveyors utilize different methods but the overall goal is determine the over all exterior **O**utside Dimension (**OD**) dimension and **I**nterior **D**imension (Interior Face of Exterior Wall to the opposite Interior Face of the Exterior Wall) or **ID** of the building.

If possible surveying and modeling the exterior envelope of a building first is preferred. Determine the over all ID (Interior Dimension) and set the correct wall thicknesses.



32'-8" - 34'-0" = 1'-4" (16"): 16"/2 = 8" Wall Thickness.

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The exterior wall thickness can also be determined with these measurements. OD – ID/ 2= Wall Thickness. The measurement is divided by two to take into account both walls. Or 32'-8" - 34'-0" = 1'-4" (16"): 16"/2 = 8" Wall Thickness. While one wall could be thicker than another it is recommended that you measure the walls manually as well to confirm.

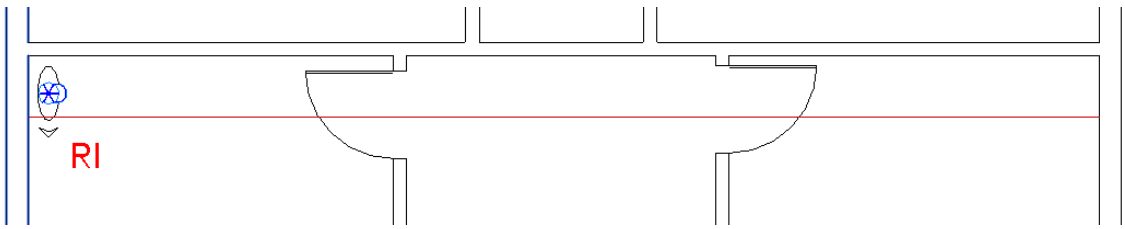
Using the Build Wall Function

Video Tutorial: Add Interior Wall Partitions to a Surveyed Wall

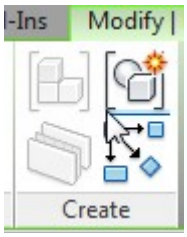
If you are starting in the interior and have the opportunity to create an entire wall length you may do so and the add the interior partition walls afterward. To do this simply measure the entire wall, and then to add interior wall partition simply 'Activate Wall" again and PKNail Pro will bring the Add Wall Dialog Box up. Instead of accepting the default, measure the location where an interior partition wall starts, enter it and start surveying.
Note: This method assumes that the entire wall follows the same line/plane.

Using Reference Lines

Using reference line serves a similar purpose as striking a line/dimension for a drawing. It allows the user to immediately see the extent of measurement, so even if you have not put a wall in immediately it can show you where your end point will eventually be. You can use the line tool in PKNail Pro. To activate the line tool Press "= ". You can then measure distance along a wall and then shoot a cross measurement R1. You can annotate the line by using the text command 'tx' or simply use different color line; the line color can be selected in the Capture Line dialog.



You will notice that there are interior wall partitions captured, you can survey them as you can encounter them and leave the R1 measurement/line in as a notation. Having the notations are useful and allow you to quickly notice any measurements that are not conforming to the building.



Note: Lines become attached to wall objects and will expand and contract if something is moved. One way to keep 'extent' lines with their proper dimensions is utilize the Group Command in Revit. Select your 'extent' lines, click the 'Group' Command. You can then relabel the Group to 'Extents' or use a naming convention accepted by your organization.

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Drafting Directly in Elevation Views

Documenting certain measurements on site, directly in an elevation view, can be critical in representing an existing building correctly. For example, documenting the height of a gutter sweep, soffit, or trim detail.

Using Reference Planes

Reference planes can be seen in all elevation views and be named accordingly so are preferred to detail lines which are view specific. These planes can represent material transitions or object placements, such as gutters, soffits, banding, etc.

Adding Column Grid Lines

Video Tutorial: Adding Grid Lines

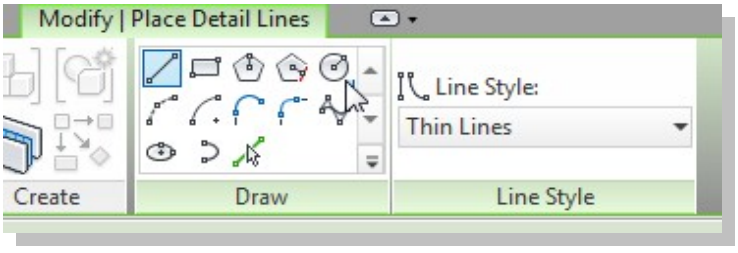
Columns are typically used to carry the load of the structure, beam, slab, etc. to the earth. Typical features of column in surveying is they appear in the same coordinates in each floor plan, and more times than not they follow a grid pattern. After surveying an entire floor plate the surveyor should have a fairly good idea of where the columns are, and what the grid is. While occasionally because of build outs around the column, the center line is not immediately discernible one can initially assume the grid by sampling the measurements between columns.

When this is done, placing column grids into the model is a good idea as it provides a reference-able grid through the project and model.

To place column lines in Revit from the Home tab scroll to the right side and find “Grid” above the “Datum” section, or type “GR”.

Mark Points by Triangulation

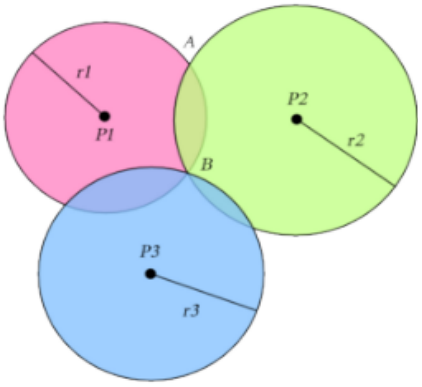
To currently mark a point via triangulation/trilateration in Revit, the easiest way to is create a radius from a known point, for example, from the end point of the wall. To do this you can type “DL” as a shortcut which brings you to the “Detail Line” menu. You



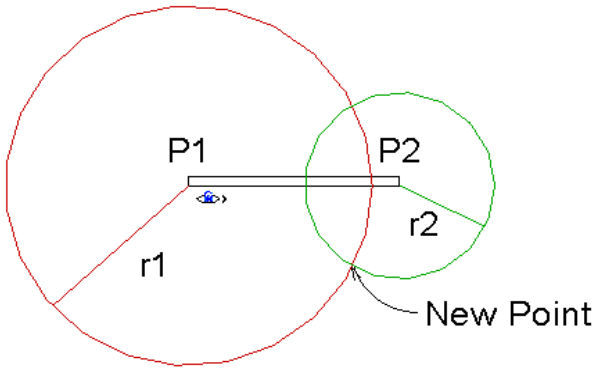
`can create a line now directly in Revit or you can choose an object on the upper right of the menu. Choose a circle, and then click on the end point of a wall you are measuring from. Enter the measured distance,

and a circle will be created with that radius. You can now create another radius from a known point, such as, the other end of the wall. The intersecting radii will mark the point you want.

PKNail Pro currently has the ability to triangulate new walls, not mark points so this technique can be helpful. In the case below, the known points P1 and P2 are the end points of the same wall.

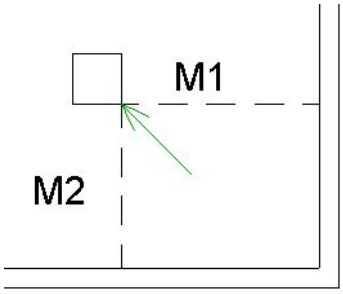


Standing at **B**, you want to know your location relative to the reference points **P1**, **P2**, and **P3**. Measuring *r1* narrows your position down to a circle. Next, measuring *r2* narrows it down to two points, **A** and **B**. A third measurement, *r3*, gives your coordinates. A fourth measurement could also be made to reduce error.



Creating Standalone Objects

While we expect to introduce workflows to take into account these issues a surveyor/modeler currently has to place stand alone objects themselves through either "marking points" as described above or creating extents from known objects. For example, if a column is located within a room, measuring to the face of two perpendicular faces will give you the location point of where the column should be placed. You can use the "DL" command to create a detail line, or use "Line Mode" to create the lines for placing objects.



Starting a Project with PKNail Pro

Connecting the Disto with your Computer/Laptop

In order to take advantage of wireless data transfer you will need to "connect" or pair the Disto with your computer. In order to do this you will need to use the Bluetooth manager available in Windows.

If you have not connected or paired your Disto with your computer/laptop you will need to do that now.



1. Make sure the Disto is charged, ready, and that you have pressed the Bluetooth Icon on the Disto Keypad.
2. Click on the Bluetooth Icon in the Tray of Windows. Typically it is in the bottom right of the computer screen next to the time and date.
3. And I the menu option to the left click "Add a Bluetooth Device" and the Set Up Wizard will start
4. Your computer will search for discoverable devices, and the Disto should show up as a device.
5. Click the Disto Icon in and Click Next. If a "passcode" is required, enter the default passcode of "0000".
6. After it is paired click the "Open Bluetooth Settings" link which should be available on the left margin in Windows 7. You will see a series of tabs, click on the "Communications Port" Tab. Within this dialog box there will be a COM Port that is assigned a direction, "Outbound", please write down the COM Port that has been assigned to "Outbound" It should look like COM5 or similar as will need that later.

*Note: Yes we want to make this easier and are working on it. Fortunately, this only has to be done once during this initial set up.

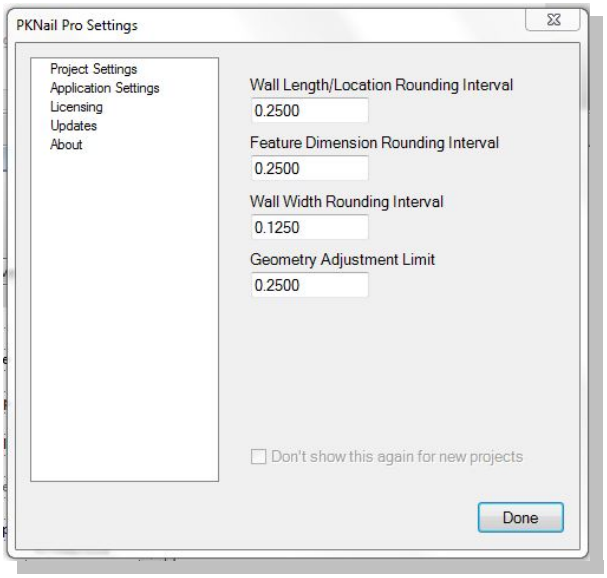
Creating a New Project Using the PKNail Pro Template

In order for PKNail Pro to operate you must use the PKNail Pro template file which was installed in the Revit Templates folder when the program was installed. To start a new project make that selection in the Revit Menu by clicking the "R" and choosing open, then choose the PKNail Pro template.

Setting Tolerances

At the beginning of a Project you should set the tolerances for PKNail Pro. You can accept the defaults or you can set them yourself. If you are going to use the same

tolerances for every project you can then check “Don’t Show This Again” option and these tolerances will remain the same for every project until you change them.



Wall Length Location

This setting will round the length of the wall. You can set it at “0” if you want or in this case it is set at .25 or ¼”. You can either notation (decimal or fraction) to enter the information. The rounded number will be the actual wall length and location will round any object you enter the wall to the same rounding interval placed here. So if you were to measure a door location from the edge of the wall and it measures 12’- 4 3/16”, PKNail Pro will place the door starting at 12’ 4 ¼” based on the .25” or ¼” rounding interval.

Feature Rounding

This will Round any feature you put in. For example, measuring a width of the window at 3’-0 1/16” will be rounded down to 3’-0”. This will happen for every measurement, Height, Width, Sill Height.

Wall Width Rounding

The default is set at .125” or 1/8”. So any wall thickness will be set on a scale of 1/8”. You can change this to whatever tolerance you like.

Geometry Adjustment Limit

This setting will allow walls to join within the set tolerance as well as build walls orthogonally as long as they are within the set tolerance. For example, when set at .25 or ¼” inch, if you measured a wall that end within a ¼” of the next wall it will automatically join them. Additionally, if when closing a wall or space, and you are within the tolerance of this setting it will close a room with an orthogonal wall rather than create a wall with an angle of 89.975 degree angle.

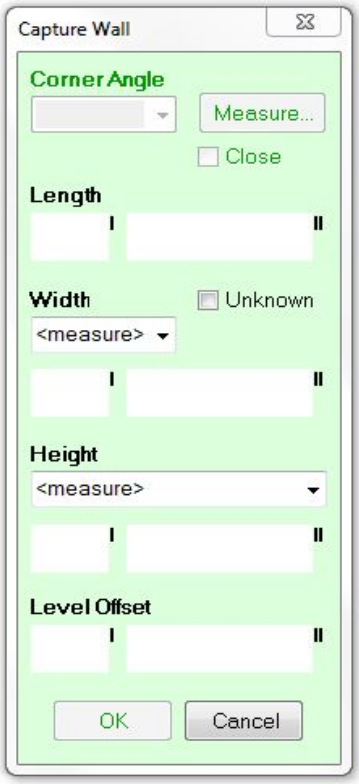
Capture Commands

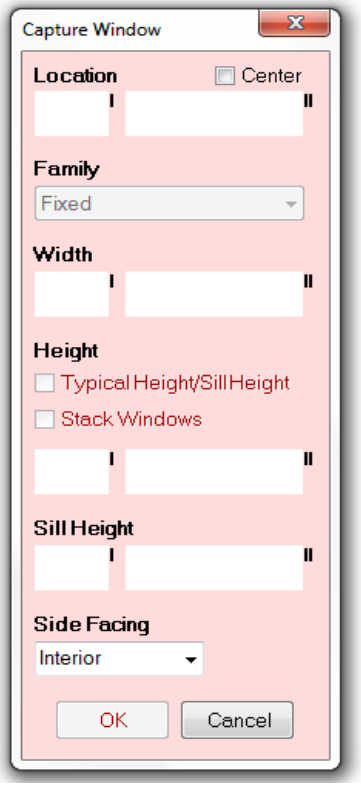
While you will be using many of the commands included with PKNail Pro we allowed the activation of the most common ones right from the range finder. Capture Wall, Capture Door, Capture Window, Finish, Cancel and Up/Down. While we will go into detail about these commands in the next sections we wanted to point out a couple of things.

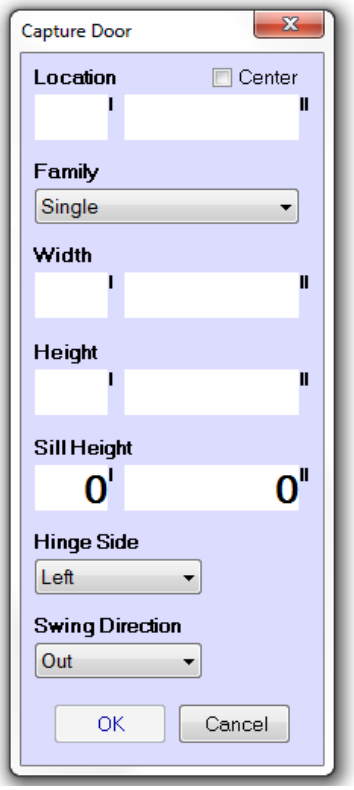
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The capture commands dialog boxes are different colors so you can distinguish them from a distance.





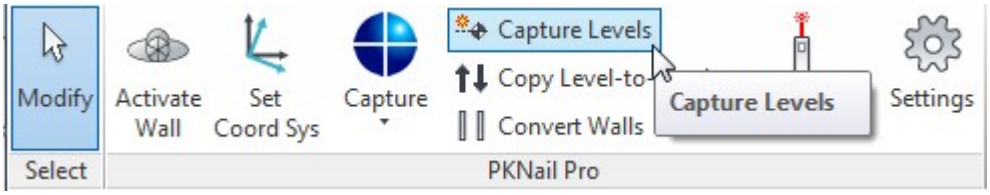


Capture Levels

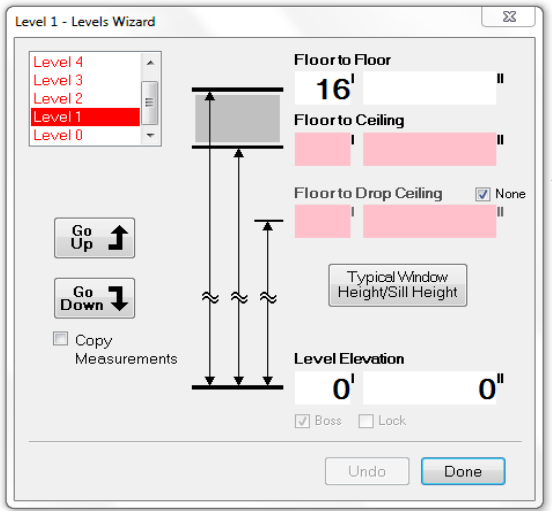
Video Tutorial: Capture Levels

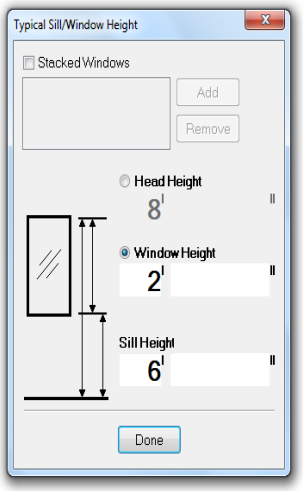
When building in 3D, Revit and BIM defining vertical data or information is critical. In the case of Revit, most elements are assigned to a level and can be offset from any one particular level. When capturing existing conditions of a building defining and capturing this information at the onset is critical for a successful survey.

First determine how many levels you are going to survey and click on the 'Capture Levels' option from the PKNail Pro ribbon:



The Levels Wizard will open with Level 1 and Level 2 already present. The default Level 1 to Level 2 is set at 10'. You will want to enter the correct information. If there is another Level, select 'Go Up', name the level and enter the correct floor to floor information, repeat this for every level you are going to survey.





Note: You can rename levels directly from the Capture Levels dialog. Just double click on the level name and type.

Adding Typical Window / Sill Height

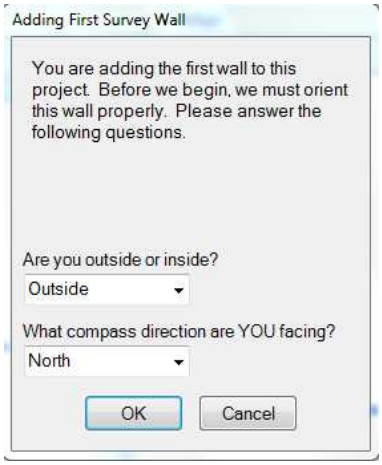
Video Tutorial: Adding Typical Windows

You can also capture the typical window height and sill height for each level you will be surveying. PKNail Pro allows for the dynamic creation of window family types based on these measurements and will help in the speed and accuracy of your surveying.

Capturing Your First Wall

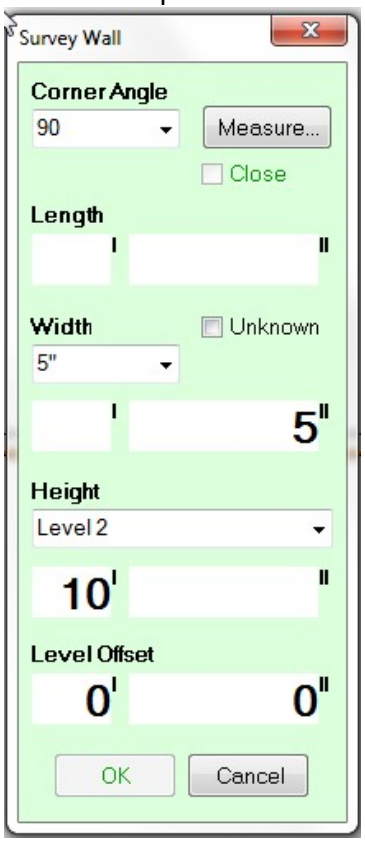
Video Tutorial: Capturing First Wall

Press the right arrow on the Disto to open the Capture Wall dialog. If the project is brand new and empty, before the Capture Wall dialog appears, you will see a dialog asking you to specify your orientation so that the project will automatically be oriented to true north:



You may enter the required information or just cancel the dialog if you don't care or plan to orient the model manually later.

Then the Capture Wall dialog will appear:



You will notice a variety of input fields awaiting data, the first of which is length. Measure the length of wall, end to end using the Disto and send the data automatically to the dialog by pressing the button.

Enter the width in one of 3 ways: either type it in manually, select it from the list of wall width types already present in the project, or measure using the Disto and send the measurement using the button. As for the height you will notice the default is set at Level 2. You can override this to be any height you want., otherwise PKNail Pro will connect the top of the wall to the level directly above the active floor plan.

Press the right cursor button twice to finish capturing and render the wall. You will notice that the default angle is set at 90 degrees. This is because the majority of walls you will encounter will be inside corners. To toggle to 270 degrees, simply press the right arrow key once after opening the Capture Wall dialog.

By measuring a length of wall and pressing the key on the Disto you can send the data wirelessly to PKNail Pro. Notice we chose the height to the next level, Level 2 and the height has been defined by data we entered previously, of course, you can override the default data if you need to.

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Double tapping the right arrow will automatically render the wall with the data contained in this dialog. In this case, a wall 50’ long, 5” wide and connected to Level 2.

Switching Survey Direction

You can survey/model in both clockwise and counterclockwise direction adding features from either end point of the wall.

 To build a wall in a clockwise; left->right direction use the right directional key.

 To build a wall counter-clockwise; right->left use the left directional key.

When referring to a directional key you will notice that there is a small blue triangle on each key pointing in a particular direction.

Adding Objects from Either Direction

If you have been surveying in a clockwise direction but then find you want to add an object taking a measurement from the right hand-side of the wall back to the object, like a window or door you have 2 methods for doing that.

Key Press Method

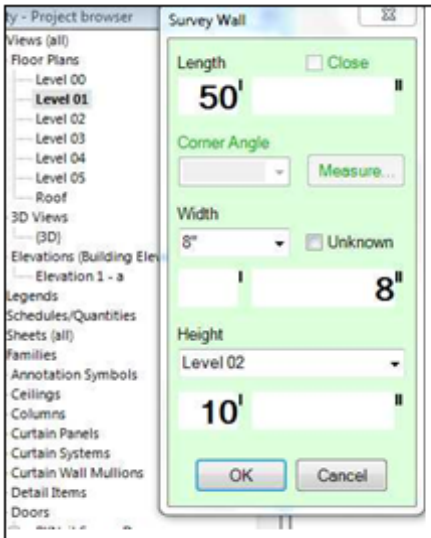
With the key press method you can simply press the object key, or the “window” or “door” key and press the corresponding directional key.

Activate Wall Method

With the activate wall method. You can simply choose to the 'Activate Wall' function and choose which side you want to add objects or start from.

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Capturing Walls





Press the Right Arrow on the Disto to open up the 'Add Wall Dialog' (See Image 1 at the beginning of this document for button description and navigation of the Disto.)

By measuring a length of wall and pressing the key on the Disto you can send the data wirelessly to our software. Notice we chose the height to the next level, Level 2 and the height has been defined by data we entered previously, of course, you can override the default data if you need to.

Double tapping the right arrow will automatically build the wall with the data contained in this dialog. In this case, a wall 50' long, 8" thick and 10' high.



When the wall is built you will note that a 'surveyor' oriented in the direction you are surveying and measuring is located on the wall. In this image his ready to measure the next feature moving left -> right, however you can add features from either direction.



Single Tap Build Wall from Left to Right
Double Tap – Builds Feature



Single Tap - Build Wall from Right to Left
Double Tap – Cancels Current Command

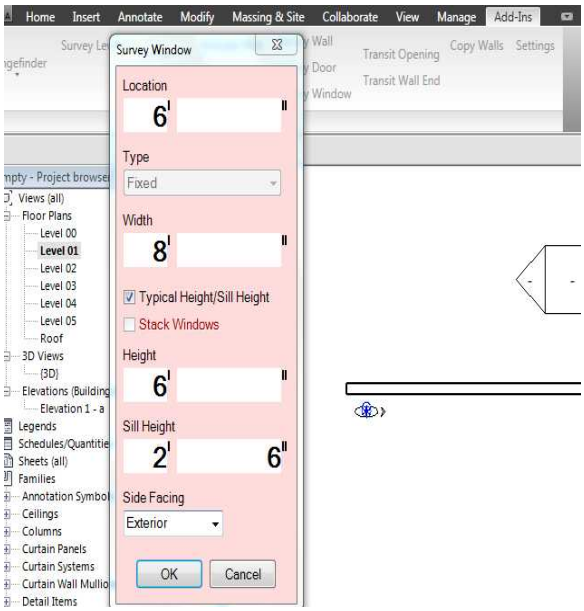


After the wall is put in, the surveyor then places architectural objects as he/she encounters them along the wall. The objects are placed via the end point of either side of the wall. Measuring from the beginning of a feature such as the window back to the basepoint and measure the width, (if using the wizard and typical height measurements the window will be placed with previously entered z axis information) otherwise measure sill height and height.

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Capturing Windows

Windows will be added relative to the direction you are measuring. For example, if you are measuring/surveying Left to Right, PKNail Pro will add the next object based on a clockwise / left to right measurement. The software will always add a feature starting at the next objects closest point to the last reference point.





To add window tap the 'plus' key. You will also notice a small blue arrow that points up and the left.

Measure the distance from the wall where the window starts and automatically send it to the program by pressing the Bluetooth button

The cursor will automatically move to the next field after entering data.

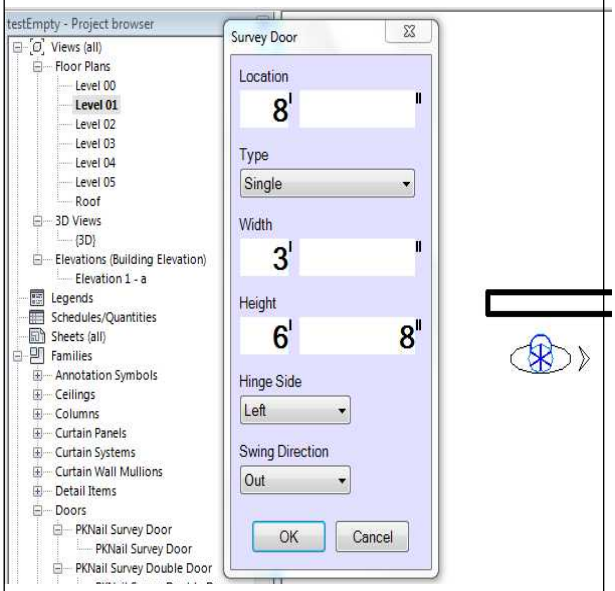
The first time measuring the width of a window this field will be blank. Otherwise, it will automatically keep the data of the last measured window. This is helpful in

To change the direction you are adding a window, press the "+" button and then the direction button.

You can then measure the width, add the width by pressing the Bluetooth button, and then double tap the right arrow key to build the window.

IF you entered Typical Height / Sill Height Information earlier PKNail Pro has already filled in that information. You can always enter different data.

Capturing Doors





To add a door tap the 'minus' key.

If you are adding the door in the direction you are already measuring, enter the distance, select the type, enter width, height and choose swing direction/hinge side and when satisfied double click the right arrow key to build. To send any piece of numerical data press the Bluetooth key.

Remember to send data, press the Bluetooth key. To build a feature, double press the right directional key.





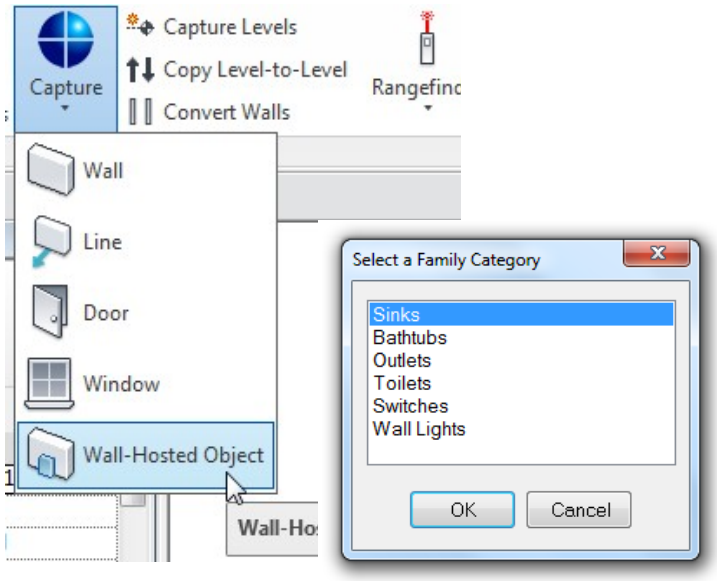
You can then measure the distance from the wall where the door starts and automatically send it to the program by pressing the Bluetooth button

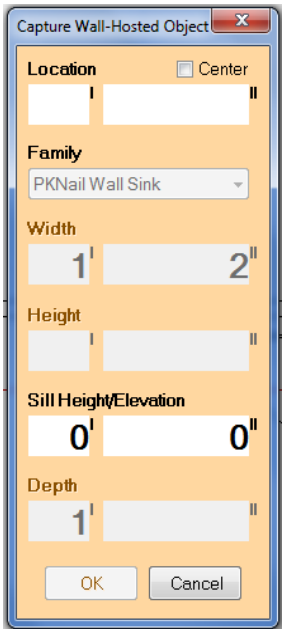
You can then measure the width, add the width by pressing the Bluetooth button, measure the height, press the Bluetooth button and then double tap the right arrow key to build the door.

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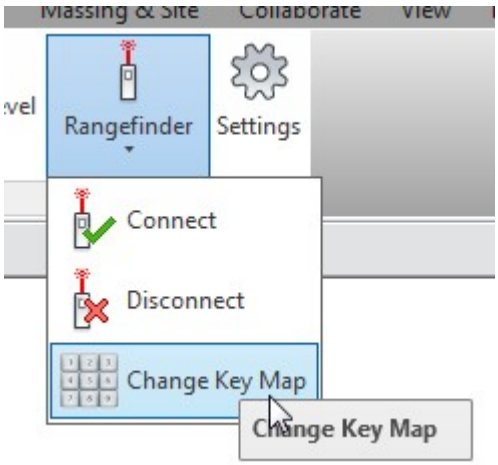
Capturing Object Instances

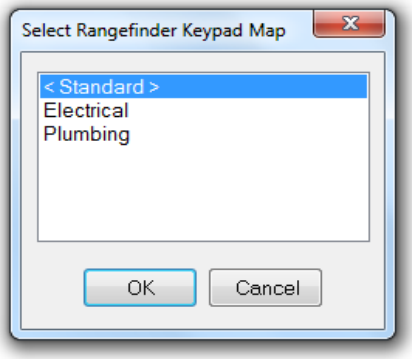
PKNail Pro gives you the ability to place wall hosted objects on any wall. You can do this in two ways. One is to select the 'Capture' Menu in PKNail Pro and select 'Wall Based Object' After clicking on the 'Wall Hosted Object' Selection, a menu of objects can be selected to place.



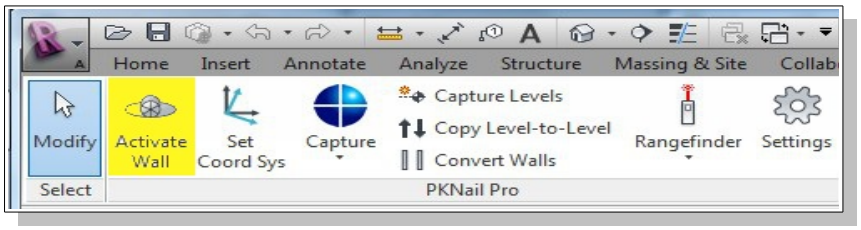


The other is to select “Rangefinder” from the PKNail Pro ribbon and change the key map. This will allow you to automatically capture Electrical, Plumbing or Fire Protection object instances directly from the Disto, same as you add walls and windows. Additionally, you can map your own custom objects directly to the Disto; see the “Customizing PKNail Pro” section for details on how to add your own families.





Activate Wall



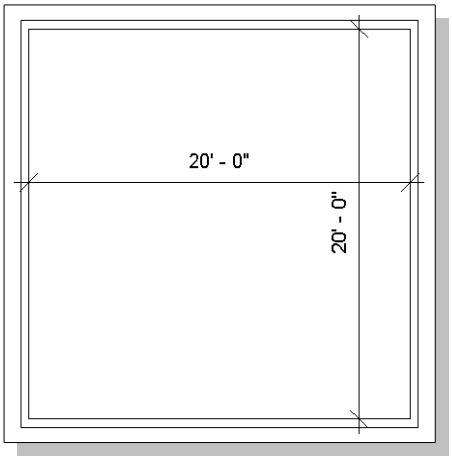
Activating a wall will allow you to move on to any wall, or switch ends of a wall. The wall can be one that was captured by PKNail Pro, drawn by hand, or from an existing model.

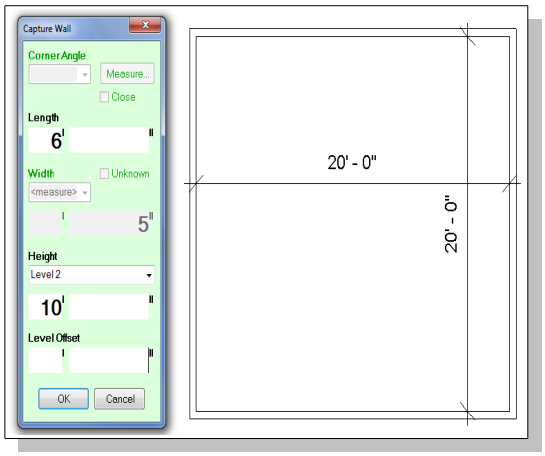
Once a wall is activated you can use PKNail Pro to add an interior partition wall, add architectural objects such as doors and windows, or add MEP objects such as Toilets, Sinks, Communication Ports and Power, or even Fire Protection objects all placed through the laser rangefinder and from accurate measured distances both in the horizontal (X,Y) and vertical (Z) planes.

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Capturing Interior Partition Walls

In some cases you may want to capture the over all dimensions of a 'larger' room or hallway and then add the interior partition walls. In this example the surveyor has created a room 20' x 20'. You then reactivate the wall with an interior partition.





Step 1: The user has created a larger room 20' x 20'

Step 2: To place 'partition' walls within the space. 'Activate' wall that contains partitions or wall where you want to 'add' something.

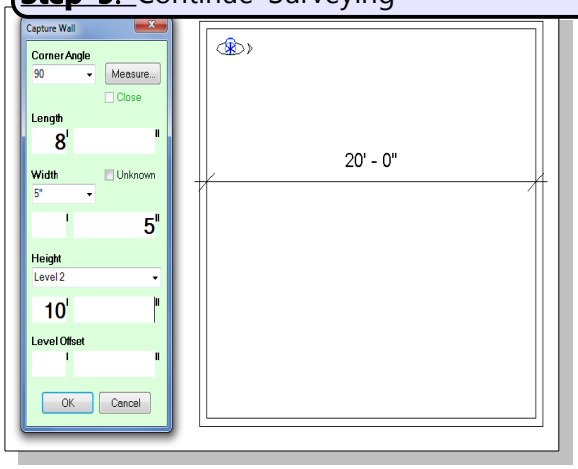
Note: The add wall dialog box will show an initial reading of the original wall measurement, in this case 20'. Override it with new measurement.

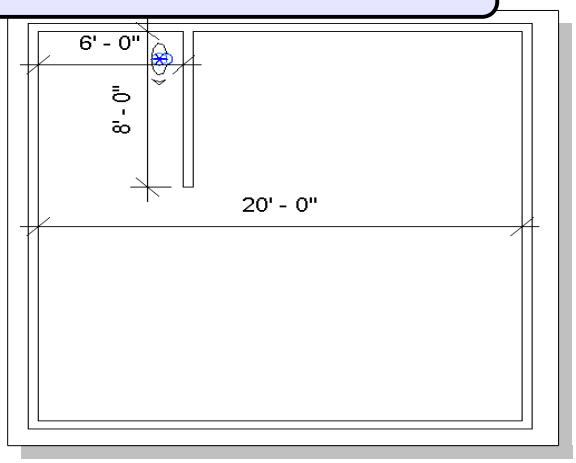
Step 3: Enter new measurement, in this case 6'-0"

Note: Notice 'surveyor' has placed themselves on the wall.

Step 4: Enter measurement of new wall. In this example, 8'-0".

Step 5: Continue Surveying



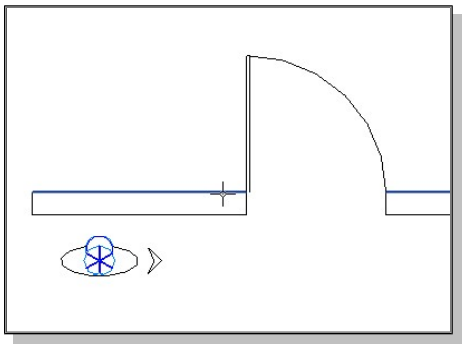


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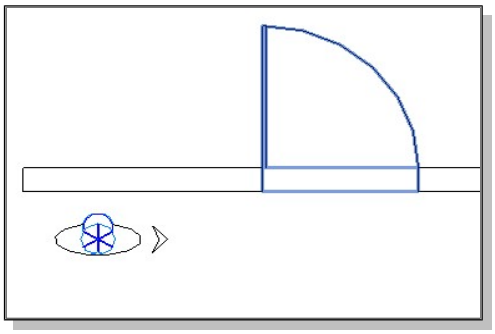
Moving Room to Room

Video Tutorial: [Activating Wall: Moving Room to Room]

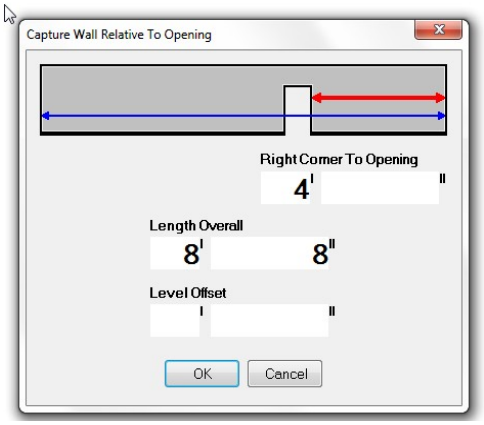
To move from one room to another use the 'Activate Wall' function from the PKNail Pro ribbon. You can click on the "other" face of the wall to activate that face. If there is a door that you are moving through, click on the "other" wall face, then click on the door or opening. You will then use one side of the door or opening as the "known reference" to tie the new room or space to the one you just captured. For example, if you select Activate Wall, then click on the "other" wall face to the right of a door (relative to the new face), then click on the door itself, PKNail Pro will use the door jamb on the right side as the reference for the new wall face, and will ask you to measure from the right jamb back to the right corner of the new space. Then PKNail Pro will ask for the overall length of the wall. You also have the option of setting a level offset for this wall face if necessary. After these are input and the dialog closed the 'surveyor' will move to the opposite side of the wall and be ready to start capturing the next space.



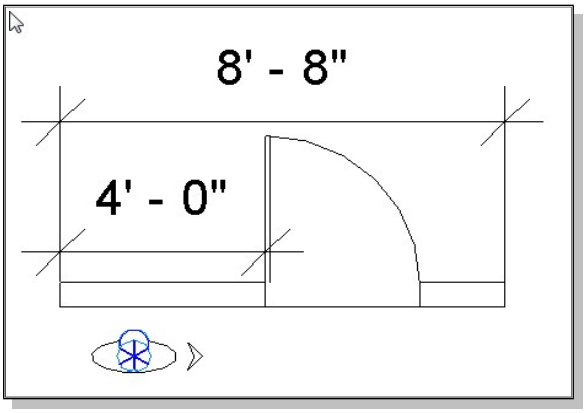
1. Choose New Wall Face



2. Select Door/Opening You Are Moving Through



3. From New Side of The Wall. Input Measurements



PKNail will default the wall end points, but any new measurements can be used

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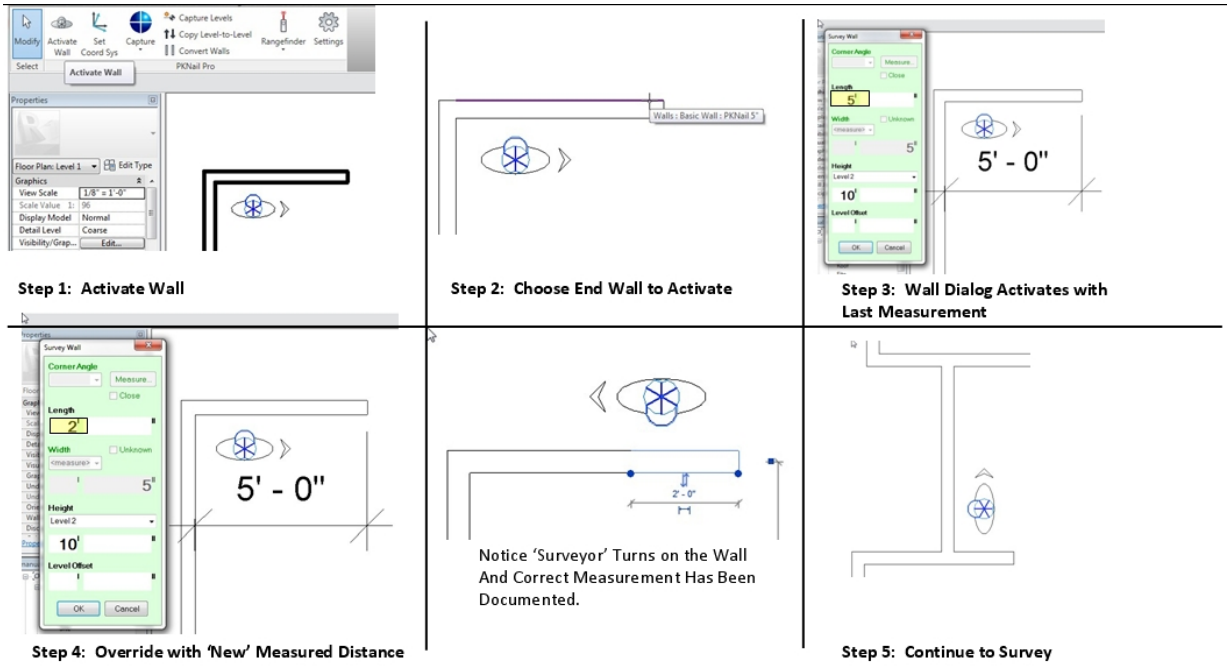
Navigating End Walls

Link: [Navigating End Wall Video]

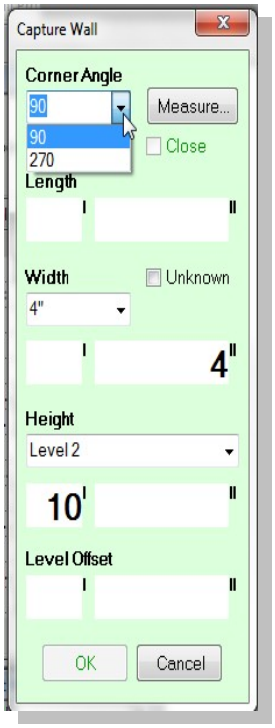
An end wall is defined in this context as a wall that has one end not connected to another wall. This happens in the case of alcoves and wing walls and you need to navigate the 'end' in order to continue surveying.

End Walls / Wing Walls / Alcoves

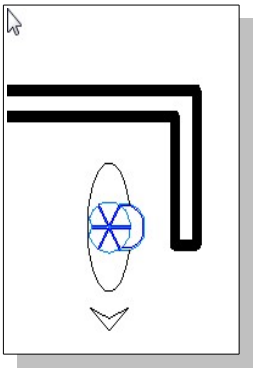
In the example below, the surveyor encounter an end wall that is 5' on one side, and on the other it measures 2 feet. In order to place the surveyor on the other side, simply use the 'Activate Wall' Function and PKNail Pro will ask every time that you want to activate a wall what the measurement is, it will default to what it 'thinks' it is which help speed surveying tremendously in most circumstances, but a user can always override this logic to input your own measurement.



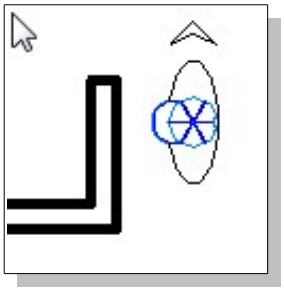
Toggling Between Inside and Outside Corners



A surveyor is routinely surveying rooms and building at 90 and 270 degree turns. PKNail Pro is set to default, or build the next wall at 90 degrees. A user can toggle this to 270 degrees by simply pressing the right  directional key.



90 Degrees



270 Degrees

Surveying Angled Walls

Video Tutorial: Surveying Angled Walls

There are two options available to you with the PKNail Pro Capture Wall dialog to capture angled walls.: “Close Room” and “Measure”.

Close Room

If you know that your other measurements are correct you can enable the ‘close room’ function in PKNail Pro. If it is “grayed” out that mean the current geometry is not allowing you to use the function. However, when active you simply have to press the right directional key once as soon as you activate the build wall dialog to enable it. PKNail Pro will then calculate the length and angle to finish and close the room. Although it says close room you can use it in exterior circumstances to close the envelope of the building.

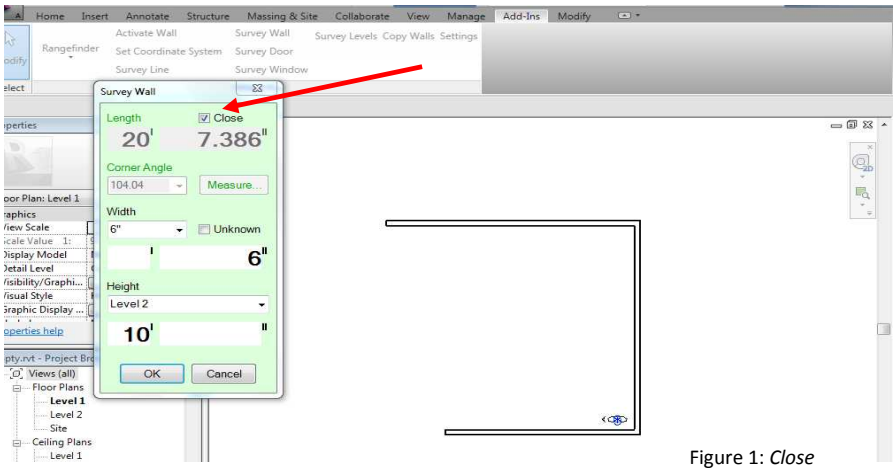


Figure 1: *Close Room*

In Figure 1: *Close Room* you can see that after the “Close” box is checked PKNail Pro has created and auto-filled the Length and Angle in order to close the room.

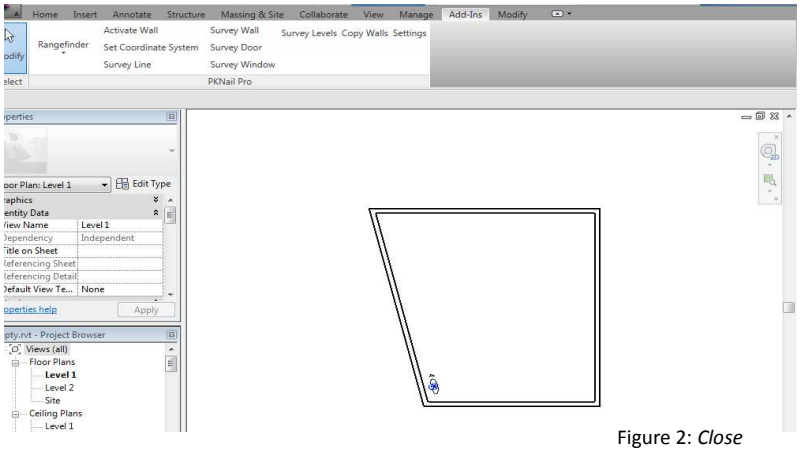
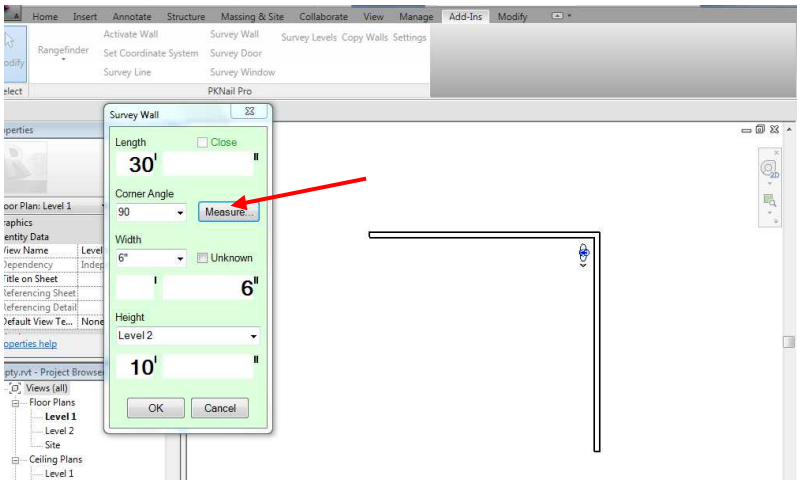


Figure 2: *Close Room*

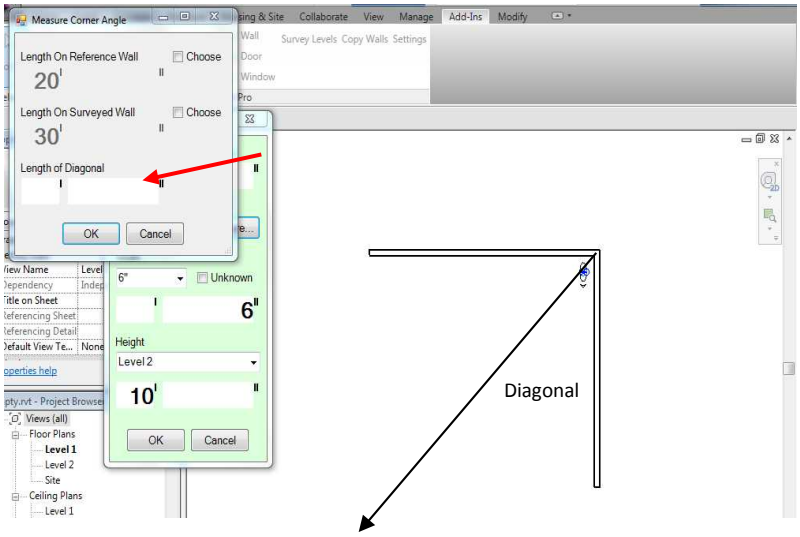
In Figure 2: *Close Room* you will see that PKNail Pro has created the wall.

Measure

Measure can be used at any time to determine an acute angle.[1] It uses the two end points of an active wall to triangulate to the end of an angled wall. First, measure the length of the actual wall. Then activate or press the measure button.

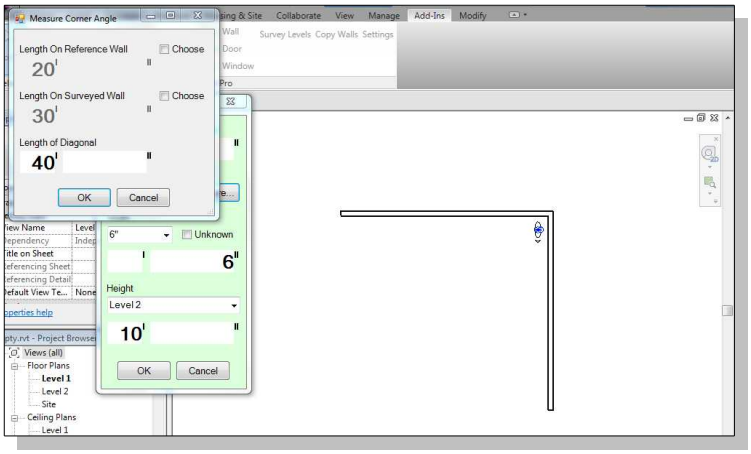


After pressing the measure button, the measure dialog button will appear, asking for the length of the diagonal. This measurement is taken from the beginning of the previous wall.

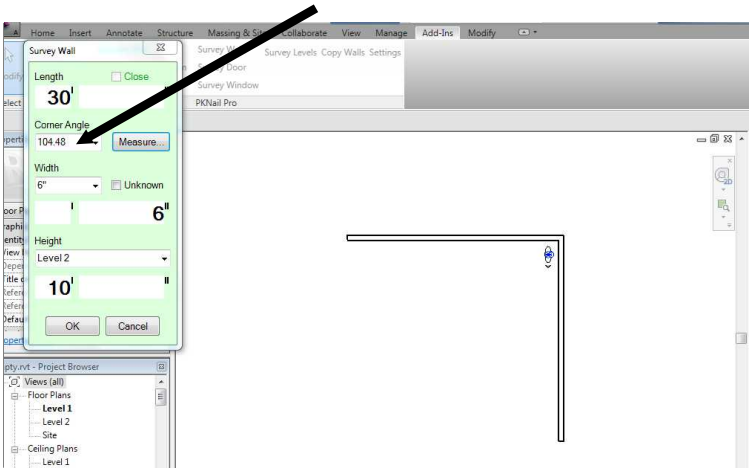


[1] In a future release PKNail Pro will be able to mark points for triangulation allowing a surveyor to mark any point within the model.

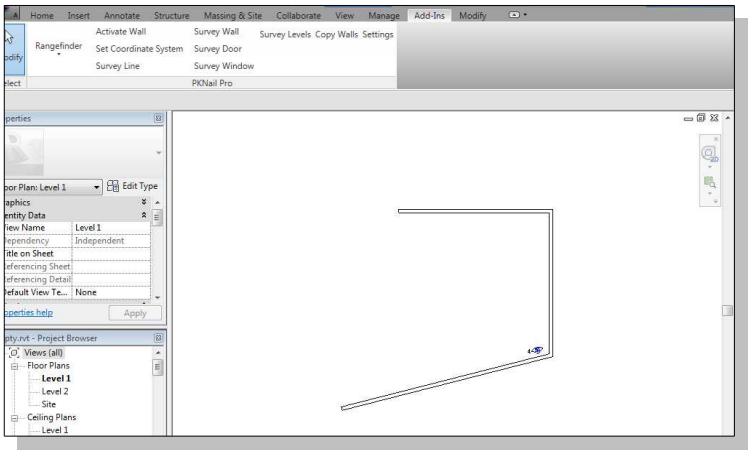
Length of Diagonal entered: Example – 40 Feet



PKNail Pro calculates the angle



The wall is captured utilizing the field information.



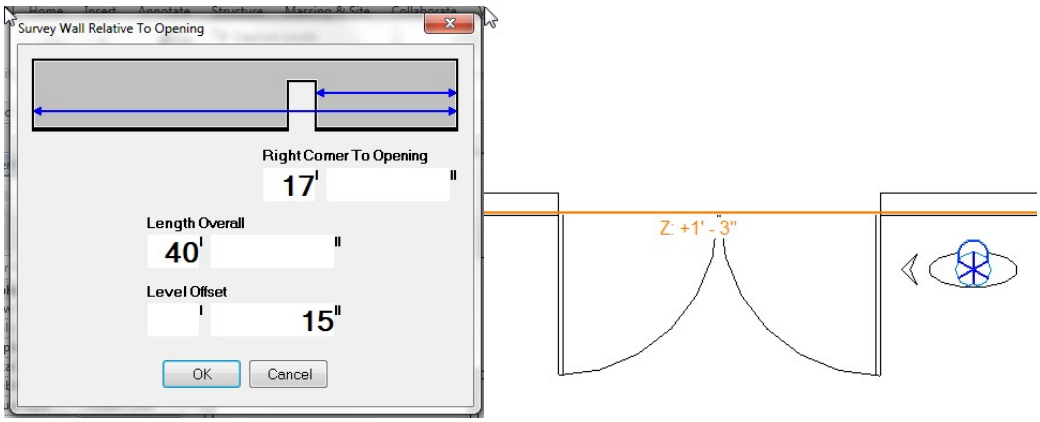
Using Level Offsets

Video Tutorial: Using Level Offsets

PKNail Pro allows you to create offsets on the 'Z' or vertical axis on a wall to include new features. You may want to include level offsets when creating a level might not make as much sense for scheduling reasons, etc. A level offset allows you to declare that any new wall features you place will be done relative to the offset. Using an offset is convenient when a new room may have a step or two up or down from the main level, or you are surveying a window on a stair landing.

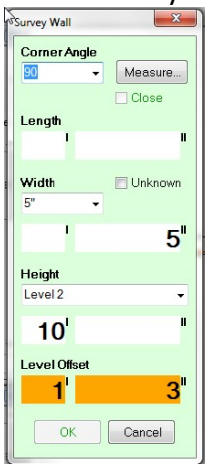
As a visual clue that you are using a level offset, PKNail Pro places an orange line and the offset distance. For example, if you are on Level 2 placing objects and declare an offset of 15", the wall will show, like the example below, an orange line with text reading Z: +1-3". Any new objects, windows, doors, electrical objects, etc. will be placed in relation to this new offset. For example, you are on a stair landing and need to place a window; instead of creating a new level for the landing, you can set the offset for the wall at the landing to 4' above Level 1. Then if you capture the window will its sill height relative to the floor of the landing, PKNail Pro will add 4' to the sill height so the window appears in the proper place on the wall. Another example is if you have a two step walk up to an office suite, etc. Using level offsets will allow you to place windows, doors, or any other object using the same PKNail Pro workflow.

When Activating a Wall



When Capturing a New Wall

When capturing a new wall where you want to use a level offset, you may enter the offset distance in the last input field of the Capture Wall dialog. When an offset is entered, the background of the input field changes to orange; this is a visual clue to remind the user they are using a level offset. Any walls captured subsequently will automatically default to the same level offset until the input field is set to zero.



As mentioned any wall that has a level offset will be annotated with an orange line and the offset distance.

Line Mode

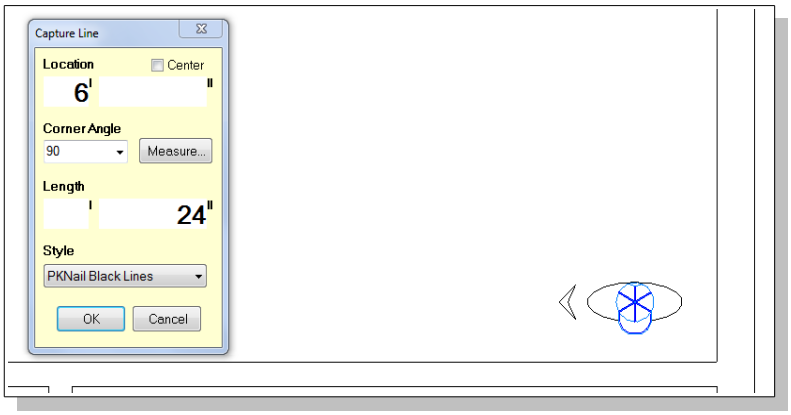
Video Tutorial: Using Line Mode *

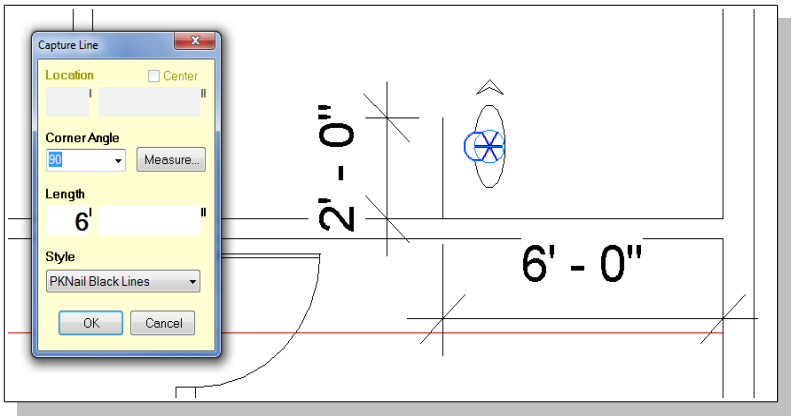


There are occasions when it is easier to create and denote spaces or objects with lines. For instance, case work and built ins. In this case you can activate Line Mode.
Line Mode can be accessed directly from the Disto by pressing the "=" Button or by selecting in the PKNail Pro ribbon.

Line Mode's first measurement is asking how far down a length of wall you want to start, and then the distance off the wall. You can then switch angles, measure the next distance or have it finish at the next wall.

PKNail Pro looks from the end of one surveying point to the next wall. So when you end at one point and look to draw another line at 90 degrees from the first, PKNail Pro will know it's 6 feet back to the wall, you can either accept this default or transfer your own measured distance.

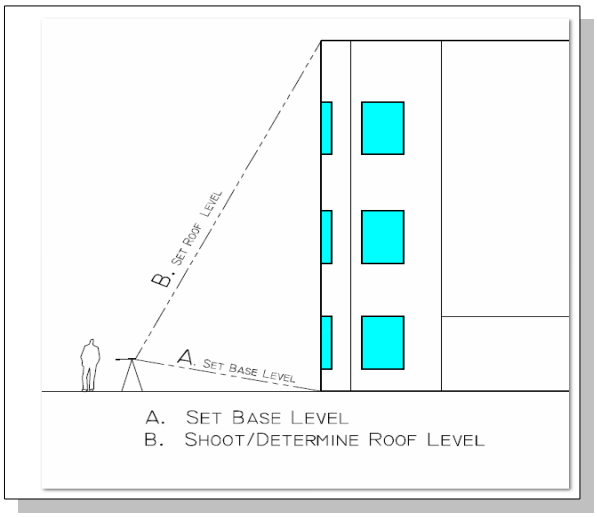




Note: Line mode can be particularly useful to capture profiles of casework and cabinetry or even locations of beams.

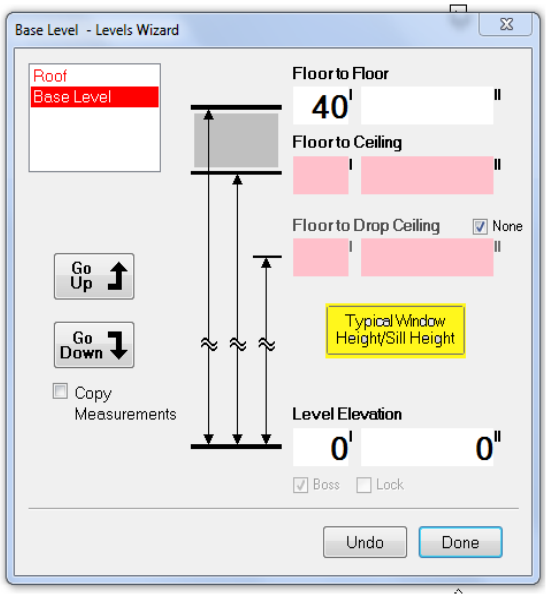
The Shell / Exterior

It is possible to survey the building and objects solely from the exterior of the building. This is helpful if you do not have access to the interior of the building or only need the exterior. In order to do this you will need to use on board Disto functions as well as the PKNail Pro. You will survey the exterior of the building the same way, but instead of surveying level to level you will be setting a base point usually the entrance level and the top point vertically, typically the roof. To get these measurements you will be using the **Finding Vertical Distances /Pythagorean Calculation 2** (Page 10) method discussed earlier in the text. You will set the Wall from the Base Level

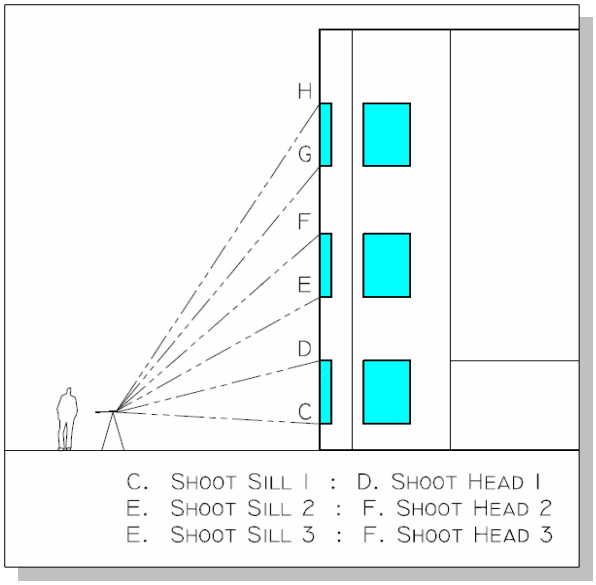


Using Stacked Windows Function

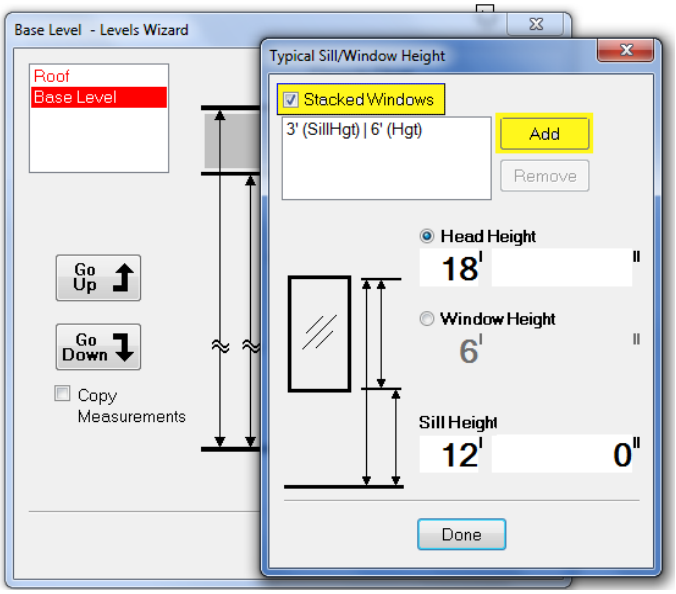
To add windows that stack vertically along the face of the building, choose 'Typical Window Height / Sill Height. Determine the measurements utilizing the **Finding Vertical Distances /Pythagorean Calculation 2** (Page 10) and enter them directly into the dialog by checking the 'Stacked Windows" function. When you have the new sill height/head height, or sill height/height combination click 'add'. This will allow any window that you add to the exterior to be stacked above it based on the information you entered here.



Step 1 (click Typical Window Height)



Step 2 (Determine Sill Height/Height: Sill Height / Head Head Height)



Step 3 (Click Stack and Add for any window that you have measured and align vertically on the building)

Adding Elevation Detail

Drafting on Elevation Views

Importing Photos into Revit

Video Tutorial: Autodesk Wiki Help: Importing Images

You can import raster images to a Revit project to use as background images or as visual aids needed during the creation of a model. By default, images are imported behind the model and annotation symbols; however, you can change the display order. You can import images into 2D views only.

You can import images using the Image tool or by dragging and dropping them from Windows® Explorer.

Video: Using Images in Revit

To import an image

Click Insert -> Click Image
In the Import Image dialog, navigate to the folder containing the image file to import. Select the file, and click Open.

The imported image displays in the drawing area and moves with the cursor. The image displays symbolically, with 2 crossing lines indicating the extents of the image. Click to place the image.

Photogrammetry

Is the practice of determining the geometric properties of objects from photographic images. Programs use algorithms to remove distortions and scale the photograph properly allowing you to use it in your modeling efforts. While this method can be find to help create representative trim details, gutter sweeps and the like if more accuracy is needed due to the scope of the project, either a total station or laser scan and its resulting pointcloud would be more beneficial.

A photo should be rectified using one the widely available third party programs before being imported to be used for detail work.

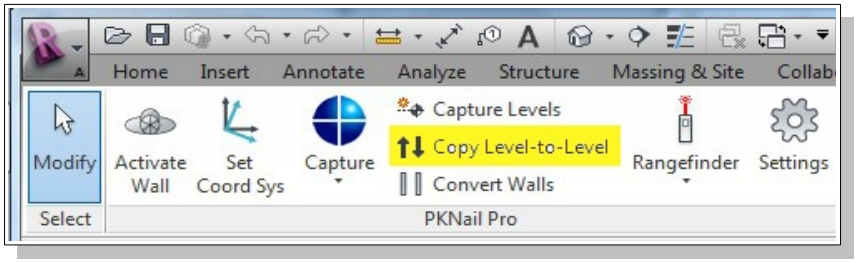
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PKNail Pro Special Feature Set

PKNail Pro has a set of features that can make the capture and modeling of existing buildings and the subsequent clean up easier. For the most part they can be accessed directly from the PKNail Pro ribbon.

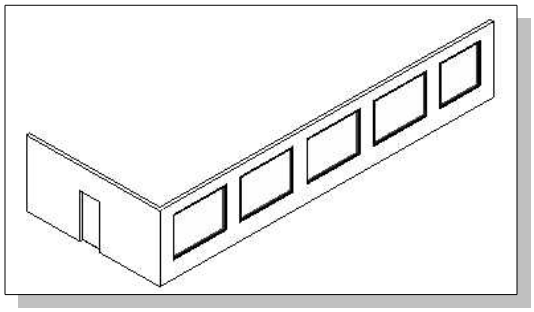
Copy Level to Level

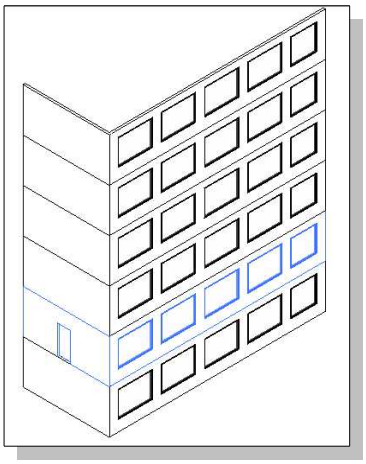
Video Tutorial: [Copying Features Level to Level]





The copy level to level is a convenience tool that allows a user to copy common features from floor to floor. Features like the exterior walls, stairwells and elevator cores can be copied, as well as objects like bathroom layouts or even Fire Extinguisher Cabinets. To Copy something Level to Level, select the features in the drawing on the Level you are working on, select where you want to copy the features to, for example, to all the levels in the building or select levels, choose whether you want all the objects or just selected and then press 'OK'.





Core and Shell

The shell of a building or building envelope can change level to level but typically will repeat itself, or certain levels may repeat themselves. After finding the building footprint through surveying with PKNail Pro or other means you can use the 'Copy Level to Level' Command to Copy objects from floor to floor. In the case of the envelope you would want to select the exterior walls that are same floor to floor as well as any 'core' walls that are similar.

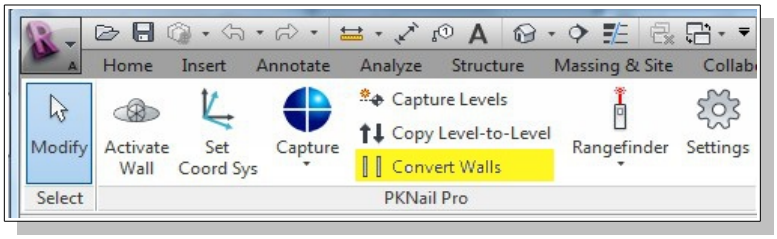
Elevator and Fire Stairs typically maintain the same vertical planes throughout the buildings. The planes, in the case of the stair well, are the stacked CMU blocks running through all the levels of the building. Additionally, the face of the elevators and the doors are typically the same. Once they have been surveyed and located within the buildings, the ‘cores’ can be copied from level to level using the ‘Copy’ Function within PKNail Pro.

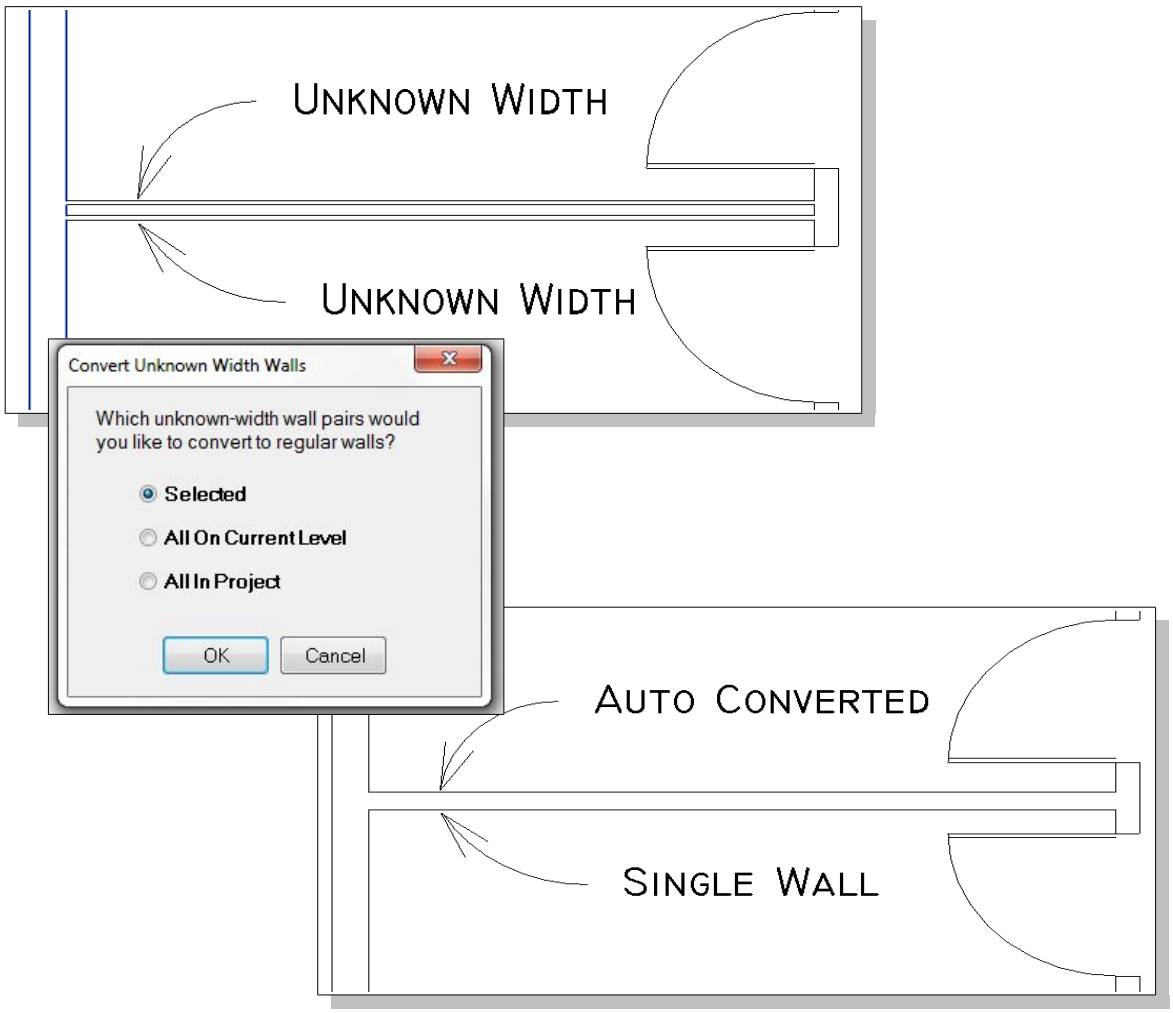
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Convert Walls

Video Tutorial: Convert Walls *

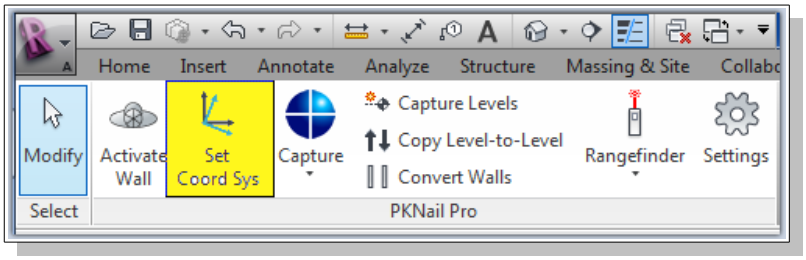
At the end of a surveying day or project you can use the 'Convert Walls' Function to convert all parallel unknown width walls to a single Revit Wall. To do this simply click the Convert Walls option in the PKNail Pro ribbon. Select either “Current Level” or “All In Project” and PKNail Pro will clean up by converting the two unknown-width walls into a single wall whose width is equal to the distance between the outside faces of the two unknown-width walls.



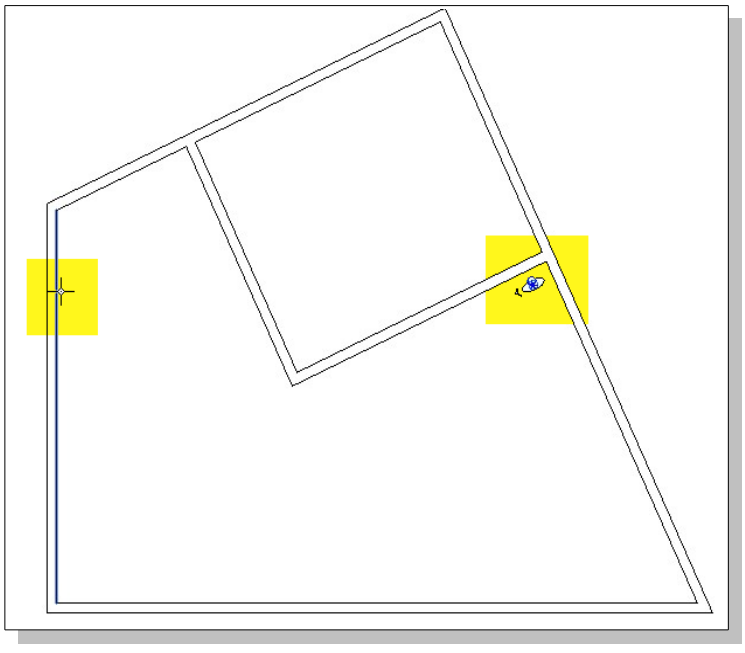


Set Coordinate System

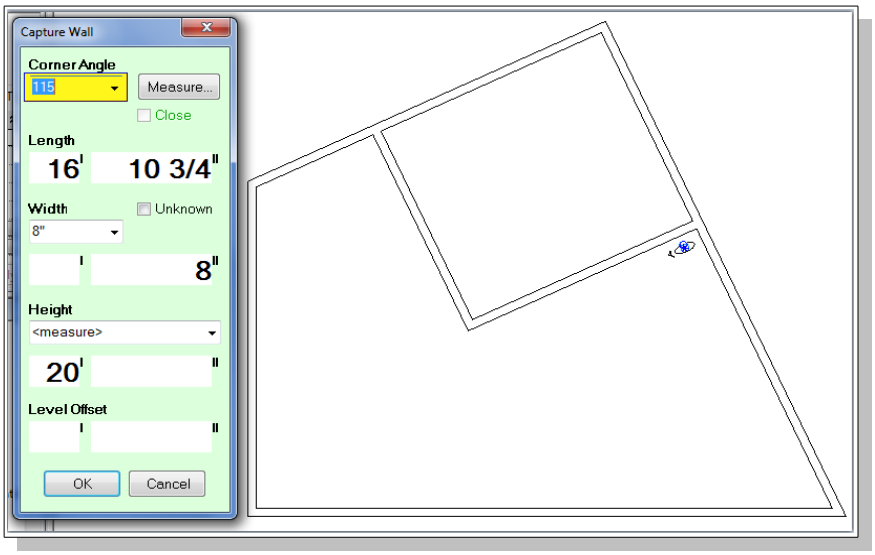
Video Tutorial: Setting Coordinate System *



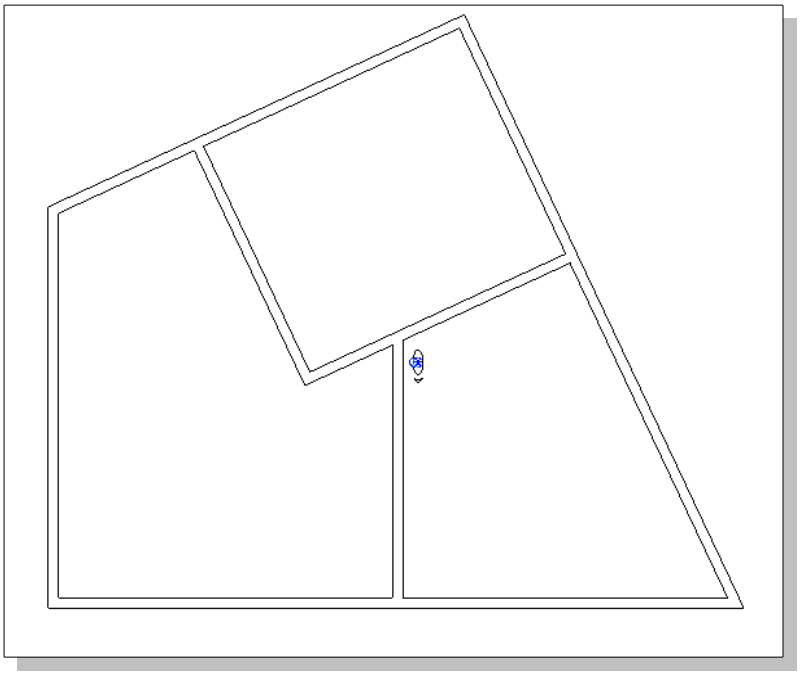
Most often any “found” angles in a building will be repeated throughout either because of a conscious design decision or because maximizing a building footprint required it to be built to a property set back that was not orthogonal. Being able to switch from a global/orthogonal setting to the new coordinate system based on the found angle is a convenient feature and one that is common. To toggle between these setting simply select 'Set Coordinate' System select a wall who's angle you want mimic with the next surveyed wall and the build the next wall.



In this example the surveyor is on an angled wall but the next wall needs to built at the global / orthogonal setting. Here the user selected “Set Coordinate System” and the clicked on the Wall who's angle is to be copied for the next surveyed wall.



You will notice that a new angle has been automatically entered in the dialog, 115°.



The new wall is built at the “New” Angle set by the user, and set by selecting the wall whose angle we wanted to copy. For AutoCad users, this works much like setting the “UCS” (universal coordinate system).

Customizing PKNail Pro

When first installed, PKNail Pro comes with a set of Revit families that have been designed to support basic reality capture, resulting in a dimensionally correct model. The generic nature of the PKNail Pro families keeps your Revit file sizes smaller and your render times faster. If your project requirements require a greater level of detail, however, you can customize PKNail Pro by changing the existing families and/or creating additional families. All of the families and parameters used by PKNail Pro are found in the Revit template used to create a new PKNail Pro project file. Using this template file as the starting point, you can create your own template files with different family combinations to support many different types of projects.

PKNail Pro has created its own way of categorizing families that is similar to but not identical to Revit categories. For each family that is used in PKNail Pro, a parameter called “PKNail Family Category” is set to either a predefined category (Walls, Doors, Windows) or a user-defined category that can be used for any other wall-hosted family. These categories are an integral part of the PKNail Pro workflow. Categories are assigned to cursor keys on the rangefinder's keypad, thus making it fast and easy to add instances of many different objects at the push of a button.

The family categories that PKNail Pro uses are: Walls, Doors, Windows, and any number of user-defined family categories. The following sections describe the level of customization available, the family parameters that are used, and the procedure for customizing or creating families for each category.

Customizing Doors, Windows and Other Wall-hosted Families

PKNail Pro can create instances of *any* wall-hosted family that exists in a project. This gives you tremendous flexibility in capturing all sorts of physical objects into your model. To make a family available to PKNail Pro, you need to do two things: follow a few guidelines in designing the family in the Revit family editor, and add some PKNail-specific parameters to the family. If you create a new PKNail Pro family category, you will also want to map this family to a cursor key on the rangefinder as well.

Family Editing Guidelines

There are a few things to keep in mind when creating or adapting families for use by PKNail Pro. First, remember that PKNail Pro is currently capable of creating instances of wall-hosted families *only* at this point in time. So if you are creating a new family from scratch, you must use the “Generic Model wall based.rft” template file. If you are adapting an existing family for use with PKNail Pro, it must either be a wall-hosted family or you will need to copy the elements from the existing family into a new wall-hosted family (again, using the “Generic Model wall based.rft” template).

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Second, make sure that family types can be created correctly by changing the Width, Height, and/or Depth type parameters. This is a basic requirement for good family design, but it it worth mentioning that these are the three parameters that PKNail Pro sets when creating new family types to match the dimensions of captured objects. The exception to this rule is if the PKNail Family Fixed Dimensions parameter is set, which disables one or more of these parameters for type creation. See below for more information about the PKNail Family Fixed Dimensions parameter.

Third, make sure you have properly set the reference planes (left-right and bottom/elevation) of your family. The vertical reference plane that is the designated left-right center of your family should be the actual center, because that will be the location of instances of the family as measured by PKNail Pro. And the horizontal reference plane that controls the elevation of the family instances should be set to the point from which you want to measure the sill height (for doors and windows) or elevation (for other wall-hosted families). For windows and doors, the reference plane should be designated as “Bottom”. For other wall-hosted families, it should be “Center (Elevation)”. For example, in the PKNail Wall Sink family, there is a reference plane that defines the top of the sink bowl. That reference plane is designated as “Center (Elevation)”. Thus, when measuring the elevation of a physical sink for capture by PKNail Pro, it should be done relative to the top of the sink bowl, which is the most convenient and accurate location.

Finally, your family must contain a “flip facing” control and should also contain a “flip hand” control. These are easy to add to a family if not already present. While in plan view in the family editor, simply select Home->Control, then select Double Vertical under Control Type, then click in the space above your family to insert the control into the view. Then select “Double Horizontal” under control type and click above or to the side of your family to insert the control. See the section below describing the parameter PKNail Family Lateral Orientation to see how the flip hand control is used to control the lateral (left-right) orientation of door instances as they are placed in the model.

Adding Family Parameters

There are a number of Revit parameters that are used by PKNail Pro to make families available for capture into a model and to control the way they are used. Some of these parameters are project parameters (defined in the project file, via Manage->Project Parameters), and the others are family parameters (defined in the family file, via Home->Properties->Family Types in the family editor).

PKNail Family Category

Parameter Name: PKNail Family Category
Parameter Type: Text
Location: Project file
Is Required?: Yes
Applicable Categories: Walls, Doors, Windows, any other wall-hosted families

Valid Values: “Walls”, “Doors”, “Windows”, “Lines”, or any name for a wall-hosted family
Description:
This parameter is the basic parameter that makes a family available to PKNail Pro, and is required to be set. There are 3 predefined values for this parameter: “Walls”, “Doors”, and “Windows” (you may also use the non-plural version, e.g. “Door”). The families that have one of these values will be available for the user to select in the relevant Capture dialog (Capture Wall, Capture Door, Capture Window). If this parameter is defined for a category and is not one of these predefined value, then it is assumed that the family is a generic wall-hosted family and will be available via the Capture Wall-hosted Object dialog. In this case, you can use any name you want, but if you want to define more than one family in the same category, the values of the PKNail Family Category parameters must be exactly the same for each family.

PKNail Family

Parameter Name: PKNail Family
Parameter Type: Text
Location: Family file
Is Required?: No
Applicable Categories: Doors, Windows, Generic wall-hosted families
Description:
This parameter gives the family an alternate name that will appear in the pick list of families in the Capture dialog. Generally you will want a shorter version of the family name for the dialog. If this parameter is not present, the name of the family will be used in the Capture dialog.
Example Usage In Default Template:
Value “Double” is used for PKNail Double door family.

PKNail Family Order

Parameter Name: PKNail Family Order
Parameter Type: Integer
Location: Project file
Is Required?: No
Applicable Categories: Doors, Windows, Generic wall-hosted families
Description:
Defines the order in which families in the same PKNail family category will appear in the pick list in the Capture dialog. This is important because it affects the rangefinder workflow-- the number of key clicks required to select the family when the Capture dialog is being driven by the rangefinder. It is desirable that the families that are encountered most often in the field be at the top of the pick list so it doesn't require as many key clicks to get to these families. If this parameter is not set, the families will appear in an undefined order in the Capture dialog.
Example Usage In Default Template:

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Since we encounter single doors most often in the field, the value for PKNail Family Order in the single door family is “1”. Thus the single door is first in the pick list and does not require any key clicks to select that family when the Capture Door dialog comes up. For double doors, it is “2”, meaning it is second in the list, and requires one key click to select in the Capture Door dialog.

PKNail Family Fixed Dimensions

Parameter Name: PKNail Family Category
Parameter Type: Text
Location: Family file
Is Required?: No
Applicable Categories: Generic wall-hosted families
Valid Values: any combination of “Width”, “Height”, “Sill Height”, “Depth”
Description:
Normally, the family parameters “Width”, “Height”, “Sill Height”, and “Depth” refer to dimensions that determine which family type should be used or created by PKNail Pro when adding an instance to the model. Sometimes we want these some of these parameters to be fixed so that every family type has the same value and we don't have to specify a value when capturing an instance. When a dimension is declared as fixed using this parameter, the corresponding field in the Capture dialog is disabled and the you won't be able to enter a value. This parameter should be a list of the dimensions which should remain fixed, separated by colons, e.g. “Width:Height”.
Example Usage In Default Template:
The PKNail Single Switch family has Width and Height parameters that determine the dimensions of instances of this family, but we don't want to have to specify these values when capturing switches, so the value of the PKNail Family Fixed Dimensions parameter is set to “Width:Height”. When this family is selected in the Capture Wall-hosted Object dialog, the width and height fields are disabled and there is no need to measure or input these dimensions.

PKNail Family Measure Location Centered

Parameter Name: PKNail Family Category
Parameter Type: Yes/No
Location: Family file
Is Required?: No
Applicable Categories: Generic wall-hosted families
Description:
Sometimes it is more desirable to capture the location of an object relative to the center of the object, rather than the left or right edge. For example, the location of a toilet is best measured relative to the center of the drain hole, rather than the edge of the tank or bowl. This parameter designates a PKNail Pro family as located via the center of the captured instance.
Example Usage In Default Template:

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The PKNail Domestic Toilet family has the PKNail Family Measure Location Centered parameter checked.

PKNail Family Lateral Orientation

Parameter Name: PKNail Family Category
Parameter Type: Text
Location: Family file
Is Required?: No
Applicable Categories: Doors
Valid Values: “None”, “Hinged”, “Sliding”
Description:
This parameter controls the wording that appears in the Capture Door dialog for door families that are not horizontally symmetrical (meaning the left side of the door does not match the right side). When the parameter is missing or set to “None”, then the door type is assumed symmetrical and the control in the Capture Door dialog for the lateral orientation is disabled. Examples of symmetrical door types are double (since there are hinges on both sides) and opening. When the parameter is set to “Hinged”, the Capture Door dialog control is labeled “Hinge Side” and the values to pick from are “Left” and “Right”. When the parameter is set to “Sliding”, the values to pick from are “LeftToRight” and “RightToLeft”. The user setting of this control determines whether the “hand” of the family instance is flipped or not.

PKNail Family Facing Orientation

Parameter Name: PKNail Family Category
Parameter Type: Text
Location: Family file
Is Required?: No
Applicable Categories: Doors, Windows
Valid Values: “None”, “Swinging”, “Facing”
Description:
This parameter controls the wording that appears in the Capture Door and Capture Window dialog for door and window families. When the parameter is missing or set to “None”, then the control is disabled in the Capture Door or Capture Window dialog. In this case the instance is placed on the wall with the default (not flipped) facing. When the parameter is set to “Swinging”, the Capture Door or Capture Window control is labeled “Swing Direction” and the values are set to “In” and “Out”. When the parameter is set to “Facing”, the control is labeled “Side Facing” and the values are set to “Interior” and “Exterior”. The user setting of this control determines whether the “facing” of the instance is flipped or not.

PKNail Family Instance Elevation

This parameter is fundamentally different from the other parameters in that it is an instance parameter whose value is set by PKNail Pro itself whenever an instance of a family is created. This feature is useful when designing families where the instances are

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likely to be created above the cut plane of the Revit floor plan view. It is common for these families to have an invisible model line extending from the bottom of the object down to the floor line. Then, when the floor plan view is rendered, the invisible model line will intersect the cut plane and the object will be rendered in the view. If this invisible model line is not present in the family and an instance is placed above the cut line, it will not show up in the plan view. As an example, if you open the standard Revit family file "Sconce Light – Flat Round.rfa", you will see the invisible model line. All of the PKNail Pro families in the electrical category have an invisible model line. The length of this line must be set properly according to the elevation of each family instance, so we use the instance parameter PKNail Family Instance Elevation to set the length of this line for each instance. If you create new families that use an invisible model line to ensure that instances are rendered in plan views, you should add PKNail Family Instance Elevation as an instance parameter to the family and then add a dimension to this line so that its length will be adjusted as instances are created.

Customizing Walls

Wall families in Revit are so-called "system families", meaning that they are built-in families that can only be created and modified in a project file. We recommend that only "generic" wall types be used by PKNail Pro, as there is no way to specify a particular wall type as you are capturing walls. The instances of these generic wall types can easily be selected (by width) and replaced by more detailed and accurate wall types later on in post-processing. The default template file installed with PKNail Pro uses generic wall types.

There are two parameters that can be set for wall types: PKNail Family Category and PKNail Wall Type Unknown Width. The following sections describe their use.

PKNail Family Category

Parameter Name: PKNail Family Category
Parameter Type: Text
Location: Project file
Is Required?: Yes
Applicable Categories: Walls
Valid Values: "Walls or "Wall"
Description:
This parameter is the basic parameter that makes a wall type available to PKNail Pro, and is required to be set.

PKNail Family Wall Type Unknown Width

Parameter Name: PKNail Family Category
Parameter Type: Yes/No
Location: Project File
Is Required?: Yes*
Applicable Categories: Walls

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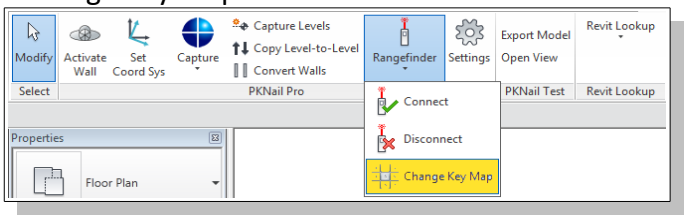
Description:
Indicates which PKNail wall type is the “unknown width” type, pairs of which can be converted into normal walls in PKNail Pro. This should be a very thin wall (in the default template, the unknown width wall type is 1 inch wide). Every PKNail Pro project must have a wall type designated as unknown width if unknown-width walls are to be captured.

Customizing Detail Lines

When capturing detail lines with PKNail Pro, you can select from a predefined set of Revit line styles, and it is easy to change the set of available line styles to suit your project or survey methodology. The set of line styles available to PKNail Pro is those line styles whose name begins with “PKNail”. To add a new PKNail Pro line style, bring up the Line Styles dialog in Revit (Manage->Additional Settings->Line Styles), then click the New button under Modify Subcategories. Make sure you put “PKNail” at the beginning of the subcategory name. Then you can adjust the line weight, color and pattern. The next time you capture a line, the new line style will be available in the pick list in the Capture Line dialog. Of course, you can alter or delete existing PKNail Pro line styles in a similar fashion using the Revit Line Styles dialog.

Customizing the Rangefinder Keypad

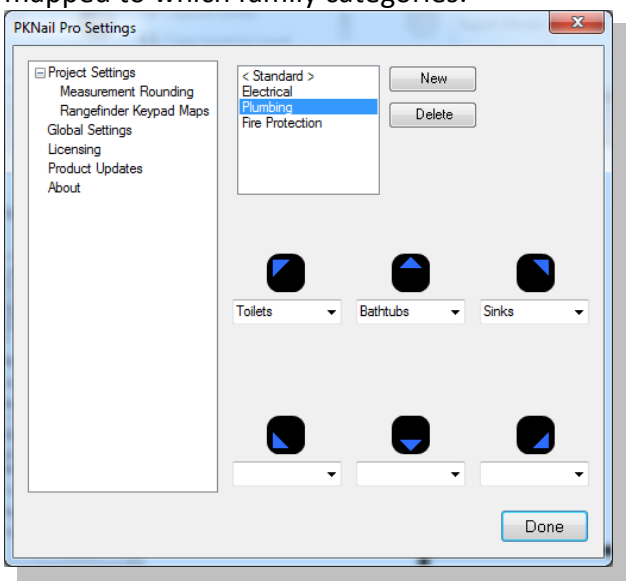
The key to PKNail Pro's speed is the ability to capture walls, doors, windows, and other objects directly from the rangefinder, without going back to interact with Revit. This is accomplished by mapping PKNail Pro family categories to keys on the rangefinder's keypad. Since there is a large number of potential categories of objects to be captured and a limited number of keys, PKNail Pro allows for the creation of multiple keypad “maps” for different purposes, in addition to the “standard” (or default) map. The default template installed with PKNail Pro contains 3 keypad maps in addition to the standard keypad map: Plumbing, Electrical, and Fire Protection. While working in the field with PKNail Pro, you can switch between keypad maps by selecting Rangefinder->Change Key Map in the PKNail Pro ribbon:



A small dialog will appear that allows you to choose the desired keypad map:

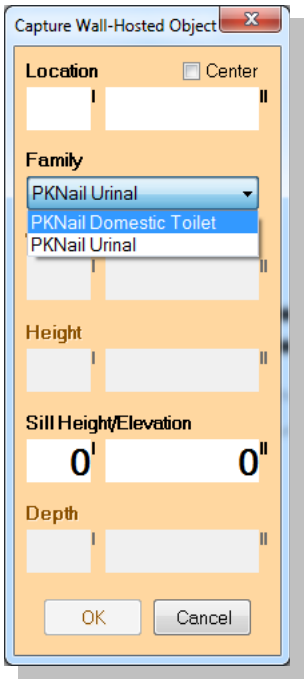


Each of these keypad maps includes keys for different family categories in the map. For example, under Plumbing, we have Sinks, Toilets, and Bathtubs. To see how the categories are mapped to rangefinder keys, select Settings from the PKNail Pro ribbon, then Rangefinder Keypad Maps under Project Settings. Then you will see a list of the keypad maps, and for the selected keypad map, a representation of which keys are mapped to which family categories:



Here we see that when the “Plumbing” keypad map is active, the up-left cursor key will capture a toilet, the up cursor key will capture a bathtub, and the up-right cursor key will capture a sink. Note that these are PKNail Pro family *categories*, not families, so there may be more than one family to choose from when the Capture Wall-Hosted Object dialog appears. For example, in the PKNail Pro template there are two families in the “Toilets” family category: “PKNail Domestic Toilet” and “PKNail Urinal”. When the Plumbing keypad map is active and you press the upper-left cursor key, the Capture Wall-Hosted Object dialog opens and you can choose between the families:

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Remember that, once you have captured an instance of one family in a category, it will be the default for that family category until the family is changed.

Combined Work Flows

Within our own practice there has been a need to utilize a variety of technologies. Most important is choosing the right one for the job; the level of detail could be very specific for a certain area, (i.e. mechanical room) and generic for others, such as hotel rooms, or office space. PKNail Pro can identify and model almost all wall types so even models created or built with native Revit can be used to start a surveying job.

With a Surveyed Shell

Some properties might have a surveyed building footprint. If you are certain of its accuracy you could import the survey data into Revit, scale it correctly, and trace over the building footprint with generic Revit walls. The width can be adjusted on site. Once on site you could take the same project, use the PKNail Pro Levels Wizard to insert the basic information, and then after determining the exterior wall thickness start surveying windows, and interior wall partitioning.

With Pointclouds

On one project due to the scope of the project, a new building was to be designed at the intersection of 2 other buildings which were built over 50 years apart and 2 completely architectural styles. We determined that the creation of a 3D database via a pointcloud and the modeling of the exterior from that data while using PKNail Pro for the interiors would be the best use of both technologies. Essentially the shell was created in the office from the Pointcloud, the interiors were created on site with PKNail Pro and the two were combined to create a unified model. A case study: Streamlining BIM for Building Documentation is available for download.

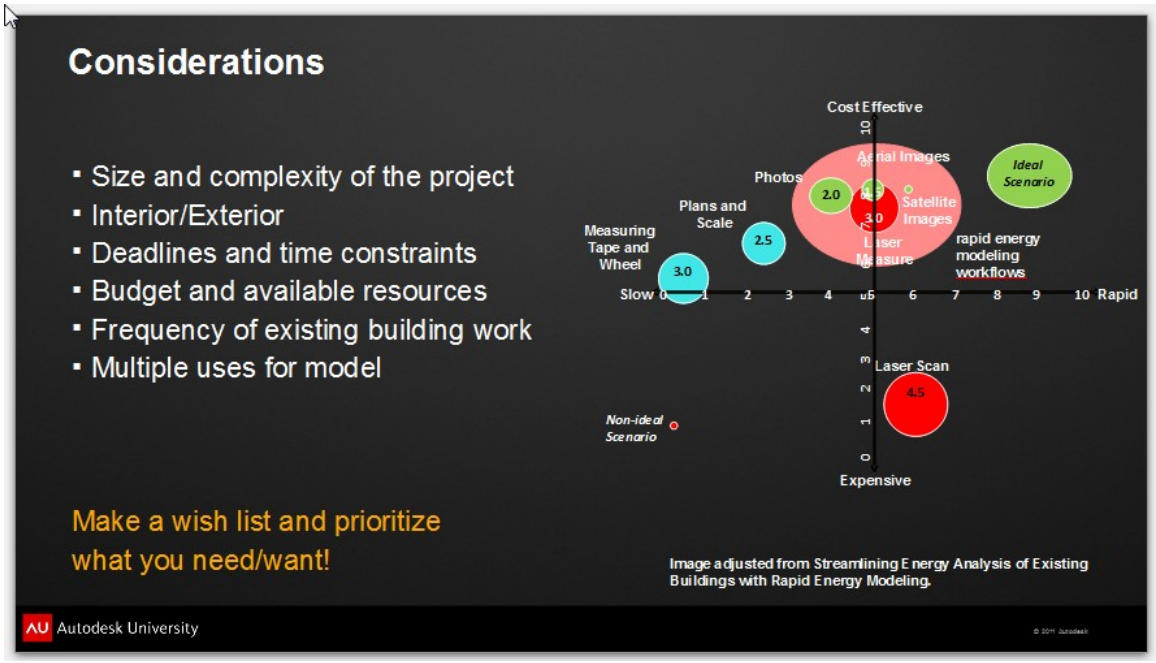
With Photos / 123D Catch

Regardless of how many photos you have you will still need some survey data in order to scale them properly. However, they can help with feature and geometry extraction. And some have used successfully for the purpose of energy modeling.

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Technology Summary

There are a variety of methods available today to document buildings; PKNail Pro, Pointclouds, Total Station. In order to best implement them on a project it is best to understand their benefits, their limits and their cost.



Courtesy of Jessica Miller: TRO Jung Brannen

Most importantly is defining the scope with the client or project manager to assess what level of detail is needed and for what purpose. While PKNail Pro can be used on many projects you can also combine it with other data capture technologies such as pointclouds. For example, you could model the building with PKNail Pro and then use pointcloud data in a mechanical room to capture all piping and other critical information if needed. Or use PKNail Pro to capture the volumes and pointclouds or a total station to capture difficult or inaccessible geometry or important details.

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Exhibit A: Scope Sheet

INTERIOR REVIEW

Survey Checklist

A safe, accurate, cost effective Survey requires a mutual understanding of "the Client(s)" requirement and scope of work.

Please mark the items on this document that are required within your scope of work:

As-Built Survey Options				
Site Information	☐	Total Square Foot of Property	☐	Number of Floors
	☐	Number of Buildings	☐	**Access Issues** - Authorization Letter and or ID Badges required
		Client Name		
		Site Address _____		
Full Site Survey	☐	Client has no CAD Drawings and requires interioreview to produce full CAD Drawings		
Cad Verification	☐	Client has CAD Drawings and requires interioreview to verify existing CAD Data	☐	Client to supply existing CAD Data prior to contract start
Cad Update	☐	Client has CAD Drawings and requires interioreview to add or update walls, structure or Assets	☐	Client to supply existing CAD Data prior to contract start
3D Revit Model	☐	interioreview to produce a 3D Revit model of the site	☐	Client to provide Revit model specifics
Survey Tolerance	☐	**interioreview** work to a survey tolerance of +/- ____"	☐	**Client preferred tolerance** +/- ____"
Layering	☐	**interioreview** work to the AIA Layering Standard	☐	Client to provide Layering Template
Sheets	☐	interioreview to supply survey on standard generic sheets	☐	Client to supply Client specific Sheet Templates
Photographs	☐	interioreview to include photographs of all internal areas of major impact	☐	External Site shots required

Scope of Work Options				
Building Floor Plan	☐	Windows (Interior & Exterior)	☐	
	☐	Stairs	☐	Attached Decks & Patios w/ Interior Access
	☐	Doors (Interior & Exterior)	☐	Columns (where visible)
	☐	Alarmed & Emergency Doors (Interior & Exterior)	☐	Column Grid (where visible)
	☐	Walls (Interior & Exterior)	☐	Flooring
	☐	Openings (Interior & Exterior)	☐	Grid to overlay on FP drawings unless otherwise specified.
	☐	Built Ins / Millwork	☐	Visible columns w/member size & frequency annotations.
	☐	Balconies		**Other**_____

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Exhibit B: Disto D8 Keypad and Features

Enabled Features for PKNail

'2nd' Function Enables PKNail Functions

Build Features Counter Clockwise
From Right → Left

Build Features Clockwise
Left → Right



1. Line Mode
2. Navigate Menu / Up

Send Data

Build Window

Build Door

Press Once - Build Wall
Press Twice – Undo Last Command

Press Once: Build Wall
Press Twice : Build*
Toggle Selection

Navigate Menu / Down

* **Press Twice** – Pressing this button twice is the same as pressing 'okay' in the dialog or 'enter'. For example, if you are in the wall dialog and you want to build a wall using the current information, just press twice and PKNail will build the wall, window, door, etc. No reason to manually hit okay or enter.
** In a menu item that has multiple choices, for example, wall angle, you can toggle between the choices (90 degrees, 270 degrees, etc)

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Exhibit C: Column Circumference to Diameter Chart

Circumference	Diameter
9.42	3
11.00	3.5
12.57	4
14.14	4.5
15.71	5
17.28	5.5
18.85	6
20.42	6.5
21.99	7
23.56	7.5
25.13	8
26.70	8.5
28.27	9
29.85	9.5
31.42	10
32.99	10.5
34.56	11
36.13	11.5
37.70	12
39.27	12.5
40.84	13
42.41	13.5
43.98	14
45.55	14.5
47.12	15
48.69	15.5
50.27	16
51.84	16.5
53.41	17
54.98	17.5
56.55	18
58.12	18.5
59.69	19
61.26	19.5
62.83	20

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Building Types

Most often building types are classified in the US in term of combustibility by the National Fire Protection Association (NFPA).. For our purposes we define them so understand what features to expect when surveying them. However, if the Building Type is used, it is the correct classification.

Type IV Heavy Timber / Bearing Masonry

Wall System Heavy Timber (Type IV) NFPA Construction is characterized by masonry walls, heavy-timber columns and beams, and heavy plank floors. These buildings have at least 12" x 12" wood beams with non combustible walls commonly brick or block construction.

Unlike ordinary construction, which typically had two bearing walls and two curtain walls, the heavy-timber building would have four bearing walls, and no curtain walls. Lintels or arches to transfer loads over openings were made from materials of substantial strength.

Occasionally, additional bracing or support cables were added to the exterior walls to provide additional strength to keep the walls from pushing outward.

Wall System

Heavy-Timber or NFPA Type IV buildings will have four bearing exterior walls made from noncombustible materials. Exterior walls typically are thicker at the lower levels, which are meant to carry the accumulated loads from the floors above. Walls (both exterior and interior) will have sockets built into them to accept floor joists or beams from the floor support system

Windows

Masonry Windows typically have a variety of components. You have the masonry opening on the exterior, and framed finish opening in the interior, providing different measurements for each. The current practice to capture these differences is to utilize PKNail Pro to capture each window/opening consistently in the building, and then to notate the details of each directly in the workstation. You can use the 'dl' detail line function to draft directly in Revit in elevation mode, or use the 'dl' function or the PKNail Pro Line function in floor plan view. While you can have custom families that have these particular parameters embedded, currently PKNail Pro supports a single parametric window type. This decision was made to minimize the complexity of data capture on site and reduce finish modeling time on site. Additional window families can be added to PKNail Pro; see the "Customizing PKNail Pro" section for instructions.

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Type V Wood Frame

Buildings include wood frame buildings with standard dimensional lumber like 2”x 4” up to and including 2” x 12” lumber.

Stick Frame / Platform Framing

Most typically stick frame construction is used.
In stick frame construction, the majority of the structural pieces are small and so the building is made up of very many of smaller pieces (i.e. sticks) placed fairly close together (typically 16” On Center) which combine to achieve the overall strength. These pieces are usually wood, but sometimes are steel. As the pieces get larger and the distance between them greater, it is no longer called stick framing, and instead becomes *post and beam* system.

Balloon Frame

Used primarily pre-1940 where floors were suspended from walls. However, in the context of creating an existing condition BIM and using generic walls types, this type of detail can be notated but will typically not change wall widths floor to floor.

Wall System

Typically plywood or a comparable sheet good (like OSB) is used to provide stiffness against wind loads, and some insulating material is filled in-between the 2x4s to give heat resisting, and then entire exterior is covered in a water resistant siding.
Traditionally, stick frame houses were made out of 2x4s, but newer houses are often made with 2x6's instead using a modified framing method often called "advanced framing" or "optimum value engineering", which is an attempt to use less lumber and allow for more insulation. Taking this concept even further, some houses are built with double 2x4 walls (in various configurations), allowing for even more insulation to be added, but at the cost of using more wood (and labor).

Recently, some builders have been substituting light gauge steel framing for 2x4 in non-load bearing walls. This can be done using continuing to use wood stick framing for the load bearing parts or going to a post and beam structure. Steel has the advantage of being recyclable (as well as typically having a high recycled content), but comes with the problem of "cold bridging" because steel is a good conductor of heat. Steel framers get around this problem by using an exterior layer of foam board insulation or, in the more energy efficient case, using a double wall.

Post and Beam

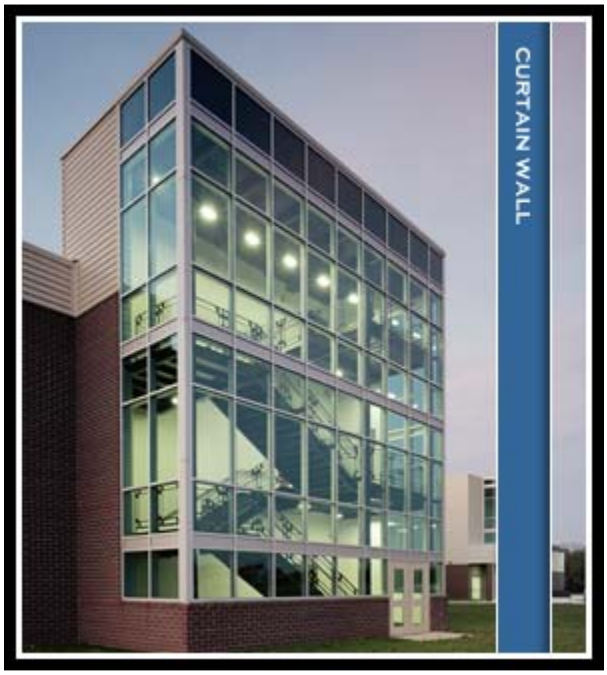
Is the method of creating structures using heavy squared off and carefully fitted and joined timbers with joints secured by large wooden pegs (larger versions of the mortise and tenon joints in furniture). It is commonplace in large barns. The methodology comes from making things out of logs and tree trunks without modern high tech saws to cut lumber from the starting material stock. Using axes, adzes and draw knives, hand

powered auger drill bits (bit and brace), and laborious woodworking, artisans or farmers could gradually assemble a building capable of bearing heavy weight without excessive use of interior space given over to vertical support posts.
In architectural terminology it can be defined as: a lattice of panels filled with a non-load bearing material or "nogging" of brick, clay, or plaster, the frame is often exposed on the outside of the building.

Steel Frame

Wall System

Curtain Walls



A curtain wall system is an outer covering of a building in which the outer walls are non-structural, but merely keep out the weather. As the curtain wall is non-structural it can be made of a lightweight material reducing construction costs.

Curtain Wall Systems are typically designed with extruded aluminum members, although the first curtain walls were made of steel. The aluminum frame is typically in filled with glass, which provides an architecturally pleasing building, as well as benefits such as daylighting. However, parameters related to solar gain control such as thermal comfort and visual comfort are more difficult to control when using highly-glazed curtain walls. Other common infills include: stone veneer, metal panels, louvers, and operable windows or vents.

Curtain walls differ from store-front systems in that they are designed to span multiple floors, and take into consideration design requirements such as: thermal expansion and contraction; building sway and movement; water diversion; and thermal efficiency for cost-effective heating, cooling, and lighting in the building.[2]

Modeling Curtain Walls

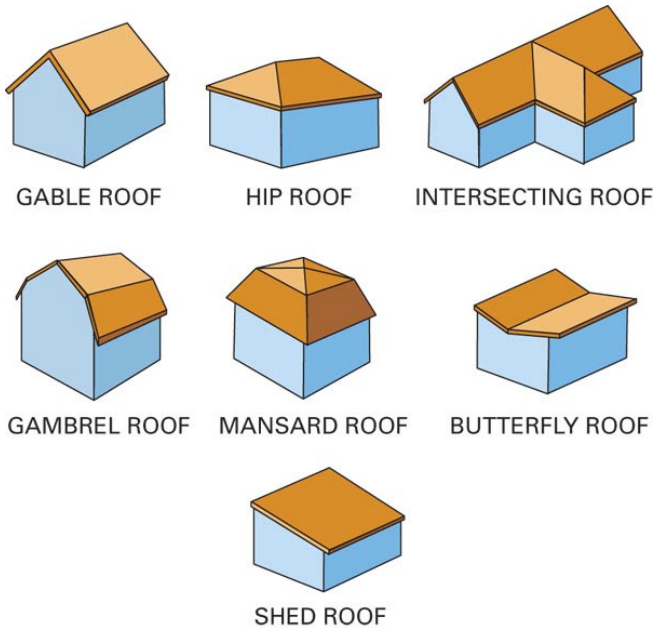
Video Tutorial: Creating Curtain Walls and Modifying Existing

[2] 'Curtain Walls': Wikipedia

Because of the custom nature of curtain walls PKNail Pro currently does not have a ‘curtain wall’ function; instead your must use a generic wall of the correct width and then place the window panels as you measure them, just as would would normal windows. This will approximate the correct curtain wall and panel, but most likely will need post processing work to convert to a true a curtain wall or to be represented correctly in elevation view.

Roof Types

While not meant to be an exhaustive list, the image bellows shows basic roof types. In the list below there are links to videos that will demonstrate how to create some of these roofs in Revit.



Gable
Cross Gable
Flat
Hip
Pyramid Hip
Cross Hipped
Hip Roof With Deck
Intersecting
By Roof Join
By Footprint
Gambrel
Mansard
Butterfly
Shed

Common Architectural Terms

BAYS The number of bays refers to the width of a building by counting the number of openings including both doors and windows. A house with a center door and a window on either side has 3 BAYS.

BELFRY A small square bell tower placed atop a roof to house a bell, often found on churches and schools.

BOARD AND BATTEN A construction method for doors or walls in which the wood is arranged in vertical boards and held in place with a horizontal board called a batten.

BRACKETS Ornamental supports, usually of wood or pressed metal, which appear at the cornice line of a building. They may be incised into a scrolled pattern or be more simply molded and are common to all Italianate style buildings, but often appear with other styles as well.

BULKHEAD A bulkhead is a set of metal door providing an outdoor entrance to the cellar.

BUTTRESS A wall support usually of stone or brick placed at the sides of a building, commonly seen on some Gothic Revival style churches.

CHAIR RAIL A chair rail is decorative wooden trim attached horizontally at the approximate height of the back of a straight chair.

CHIMNEYS Chimneys are usually built of stone or brick (more modern chimneys may be of cinder block) and are located at either the exterior side walls of the building or at the center or interior of the building. Certain vernacular folk building patterns locate the chimney at the center of the house or at the corner.

CLAPBOARD A narrow wooden board, thinner at one edge than the other, applied horizontally to the exterior walls of buildings to form a weather-tight wall surface.

COIGNS An exterior angle of a wall or other piece of masonry; usually referenced when the stones used in forming such an angle, often being of larger size and dressed or arranged so as to form a decorative contrast with the adjoining wall

COLUMN A support pillar, usually round, found on porches and as a decorative detail.

COLUMN CAPITALS Capitals are the tops of round columns and may be of several distinct types or orders. Greek Doric capitals are fluted and plain, Roman Doric capitals are smooth and plain, Ionic capitals have a rams horns at all four corners, and a Corinthian capital is highly decorative with curling acanthus leaves.

COPING The capping at the top of a wall for protection from weather elements.

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CORBEL A decorative use of brick atop the windows, walls or chimney or to create the shape of a bracket or dentil at the top of a building beneath the cornice.

CORNICE A cornice is the finished edge of the roof where it meets the exterior wall, of varying sizes, sometime plain, but often decorative and marked by brackets, dentils, medallions or some other decorative feature.

CRENALATED PARAPET A low retaining wall at the edge of a roof or porch with a uniform pattern of openings creating a battlement. In medieval times the openings were used for the defense of fortresses, hence the term battlement.

CRESTING Roof cresting is a lacy decorative fencing made of wrought iron, rimming the edge or peak of a roof, often seen in Second Empire (Mansard) style buildings.

CUPOLA A cupola is a decorative, small, projecting tower at the top of the roof of a building, often square, round or octagonal in shape.

DORMER A window opening at the roof level, topped by a front gable or shed roof.

EAVES The edge of the roof that overhangs the exterior walls, sometimes with exposed rafters.

EYELID DORMER A half-elliptical decorative window placed in the roof surface, resembling the shape of an eye.

FACADE The face of a building, usually referring to the front.

FANLIGHT A semi-circular (fan shaped) window placed atop a door, commonly seen in Federal and Colonial Revival style buildings.

FENESTRATION PATTERN The arrangement of windows across the facade of a building.

FINIAL A decorative piece set atop a spire, cupola, gable or gate post.

FLEMISH GABLE A decorative gable form, often seen in Flanders and the Netherlands, the sides of which drop in a cascade of right angles, also called a crow-stepped gable. Used as a decorative embellishment in Victorian era styles in the USA.

FLOOR PLAN The layout of the various levels of a building, showing the location of rooms, interior walls, chimneys, porches and staircases.

FLUTING Fluting is a decorative finish for wooden columns or trim where parallel grooves are carved vertically along the surface.

FRIEZE A frieze is the panel beneath the cornice at the top of a building' exterior wall which is often ornamented with brackets, dentils or medallions.

LINTEL The flat horizontal piece at the top of a window.

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MASONRY A type of construction using stone, brick, tile or concrete block using mortar.

MOLDING A decorative raised surface along the edge of an architectural feature such as a window, column, door or wall.

MORTAR A mixture of sand, water, lime and cement used to lay bricks, stone, tile or concrete block.

MULLIONS The wooden divisions between panes of glass on windows.

OGEE ARCH A center pointed arch with reverse curve sides, often seen on Exotic Moorish Revival style buildings **.**

ORIEL WINDOW A projecting bay window supported by brackets or a triangular support piece.

PALLADIAN WINDOW A three-part, round-arched window, named for the 15th century Italian architect Andreas Palladino, also known as a Venetian Window and common in the Georgian and Colonial Revival styles.

PARAPET A parapet is a low stone or brick wall at the top of a building. A crenelated parapet has rhythmic breaks in the wall to create a pattern of battlements.

PEDIMENT A triangular space created by a front facing gable roof, often seen in Classical Revival style buildings.

PENDANT An ornamental piece of wood or metal hanging down from a porch, cornice or bracket.

PENT ROOF A narrow shed style roof placed above the first floor of a building to protect the doors, windows and lower walls, often covering all four sides of the building.

PILAR A support column without classical detailing.

PILASTER A pilaster is a narrowly protruding column attached to a wall, giving the illusion of a real free standing support column.

PLENUM In building construction, a plenum (pronounced PLEH-nuhm, from Latin meaning *full*) is a separate space provided for air circulation for heating, ventilation, and air-conditioning (sometimes referred to as HVAC) and typically provided in the space between the structural ceiling and a drop-down ceiling. A plenum may also be under a raised floor. In buildings with computer installations, the plenum space is often used to house connecting communication cables. Because ordinary cable introduces a toxic hazard in the event of fire, special plenum cabling is required in plenum areas.

POINTED ARCH An arch with a strong center point, usually seen in Gothic Revival style buildings.

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PORCH A roofed space outside the mains support walls of a building.

PORTICO A small entrance porch.

QUOINS Quoins are decorative rectangles or squares of stone, brick, wood or concrete, placed at the corners of buildings to add architectural interest.

RAFTERS The wooden structural support beams for a roof, sometimes visible on the exterior for certain building types and styles.

ROOF Roofs can be steep, flat or gently sloped and take many forms, gable, gambrel, hipped, stepped gable, shed, pent or Mansard. The roof type is an important key to identifying the style of a building.

ROUND ARCH A semicircular arch over a window or door.

SEGMENTAL ARCH A slightly rounded arch over a window or door.

SEMI-ELLIPTICAL ARCH A elongated round arch over a window or door.

SILL The flat horizontal bottom piece of a window or door, often of wood, but sometimes of stone.

STOOP The uncovered wide step leading into the front or main door of a building.

STORIES The number of stories a building reflects its height by counting the stacked floors. If a building has dormer windows inset into the roof, that top section of the building is called a 1/2 story.

STUCCO A thin coating of plaster applied over exterior walls.

TOURELLE A small tower, often trimmed with corbelling.

TOWER A tall structure, either square or round in shape, rising higher than the rest of the building.

TRACERY WINDOW A pointed arch window filled with curving stone mullions often seen on Gothic Revival style buildings.

TRANSOM LIGHT A flat, glass panel above a door, usually multi-paned.

TUDOR ARCH A flattened arch with a center point above a door or window, commonly seen in Tudor Revival style buildings, (also called a 4 centered arch).

TURRET A small tower at the corner of a building.

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WAINSCOT The wainscot is the wood covered lower portion of an interior wall, usually topped by a chair rail. A wooden wainscot can be plain or paneled with a pattern of raised wooden trim.

WALLS Historic exterior wall construction can be of log, stone, brick, frame or stucco over such. In the more modern era, wall material could be of formed concrete, glass, or metal. Carrera glass was an early 20th century innovation producing a sleek, smooth colored glass finish, often applied to first floor walls of commercial buildings.

WEATHERBOARD An exterior horizontal wooden board applied with the lower edge overlapping the board below used to form exterior walls (wider and less shaped than a clapboard, although used for the same purpose).

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